

Tutorial 12: nMDS, PERMANOVA and SIMPER

ENVX2001 – Applied Statistical Methods

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Semester 1

Aim of this tutorial

In the previous tutorial we used **hierarchical clustering** to ask whether sites with similar bird communities group together.

In this tutorial we use the **same Mac Nally bird data**, but analyse it using three related multivariate methods:

1. **nMDS** – to visualise similarity among bird communities.
2. **ANOSIM** – to test whether communities differ among habitat groups.
3. **PERMANOVA** – to test whether habitat type explains multivariate differences in bird assemblages.
4. **SIMPER** – to identify which bird species contribute most to differences among habitat groups.

The data are based on Mac Nally's study of forest and woodland birds along a 250 km transect in south-eastern Australia. The paper examined bird assemblages across contrasting forest and woodland habitats, including river red gum/grey box woodlands, foothill forests, montane forests, red ironbark forests, manna gum woodlands and coastal banksia woodland.

Mac Nally used multivariate approaches to understand habitat structure, bird occurrence and abundance across sites. Here, we focus on the **bird community matrix**: rows are sites and columns are bird species.

Load packages

```
CODE
```

```
library(vegan)
```

OUTPUT

```
Loading required package: permute
```

CODE

```
library(MASS)
library(ggplot2)
library(dplyr)
```

OUTPUT

```
Attaching package: 'dplyr'
```

OUTPUT

```
The following object is masked from 'package:MASS':

  select
```

OUTPUT

```
The following objects are masked from 'package:stats':

  filter, lag
```

OUTPUT

```
The following objects are masked from 'package:base':

  intersect, setdiff, setequal, union
```

Load the data

The dataset contains:

- SITE = site name
- HABITAT = habitat category
- bird species columns = abundance/density estimates for each bird species

CODE

```
Birds <- read.csv("data/macnally.csv", header = TRUE)

head(Birds)
```

OUTPUT

	SITE		HABITAT	V16ST	V2EYR	V36F	V4BTH	V5GWH	V6WTTR	V7WEHE	V8WNHE
1	Reedy Lake		Mixed	3.4	0.0	0.0	0.0	0.0	0.0	0.0	11.9
2	Pearcedale	Gippsland Ma		3.4	9.2	0.0	0.0	0.0	0.0	0.0	11.5
3	Warneet	Gippsland Ma		8.4	3.8	0.7	2.8	0.0	0.0	10.7	12.3
4	Cranbourne	Gippsland Ma		3.0	5.0	0.0	5.0	2.0	0.0	3.0	10.0
5	Lysterfield		Mixed	5.6	5.6	12.9	12.2	9.5	2.1	7.9	28.6
6	Red Hill		Mixed	8.1	4.1	10.9	24.5	5.6	6.7	9.4	6.7
V9SFW V10WBSW V11CR V12LK V13RWB V14AUR V15STTH V16LR V17WPHE V18YTH V19ER											
1	0.4	0.0	1.1	3.8	9.7	0.0	0.0	4.8	27.3	0	5.1
2	8.3	12.6	0.0	0.5	11.6	0.0	0.0	3.7	27.6	0	2.7
3	4.9	10.7	0.0	1.9	16.6	2.3	2.8	5.5	27.5	0	5.3
4	6.9	12.0	0.0	2.0	11.0	1.5	0.0	11.0	20.0	0	2.1
5	9.2	5.0	19.1	3.6	5.7	8.8	7.0	1.6	0.0	0	1.4
6	0.0	8.9	12.1	6.7	2.7	0.0	16.8	3.4	0.0	0	2.2
V20PCU V21ESP V22SCR V23RBT V24BFCS V25WAG V26WWCH V27NHHE V28VS V29CST											
1	0	0.0	0.0	0.0	0.6	1.9	0	0.0	0.0	1.7	
2	0	3.7	0.0	1.1	1.1	3.4	0	6.9	0.0	0.9	
3	0	0.0	0.0	0.0	1.5	2.1	0	3.0	0.0	1.5	
4	0	2.0	0.0	5.0	1.4	3.4	0	32.0	0.0	1.4	
5	0	3.5	0.7	0.0	2.7	0.0	0	6.4	0.0	0.0	
6	0	3.4	0.0	0.7	2.0	0.0	0	2.2	5.4	0.0	
V30BTR V31AMAG V32SCC V33RWH V34WSW V35STP V36YFHE V37WHIP V38GAL V39FHE											
1	12.5	8.6	12.5	0.6	0.0	4.8	0.0	0	4.8	26.2	
2	0.0	0.0	0.0	2.3	5.7	0.0	1.1	0	0.0	0.0	
3	0.0	0.0	0.0	1.4	24.3	3.1	11.7	0	0.0	0.0	
4	0.0	0.0	0.0	0.0	10.0	4.0	0.0	0	2.8	0.0	
5	0.0	0.0	0.0	7.0	0.0	0.0	6.1	0	0.0	0.0	
6	0.0	0.0	0.0	6.8	0.0	0.0	0.0	0	0.0	0.0	
V40BRTH V41SPP V42SIL V43GCU V44MUSK V45MGLK V46BHHE V47RFC V48YTBG V49LYRE											
1	0	0.0	0.0	0.0	13.1	1.7	1.1	0	0	0	0
2	0	1.1	0.0	0.0	0.0	0.0	0.0	0	0	0	0
3	0	4.6	0.0	0.0	0.0	0.0	0.0	0	0	0	0
4	0	0.8	0.0	0.0	0.0	1.4	0.0	0	0	0	0
5	0	5.4	0.0	0.0	0.0	0.0	0.0	0	0	0	0
6	0	3.4	2.7	1.4	0.0	0.0	0.0	0	0	0	0
V50CHE V51OWH V52TRM V53MB V54STHR V55LHE V56FTC V57PINK V580BO V59YR V60LFB											
1	0	0	15	0	0	0	0.0	0	0.0	0	2.9
2	0	0	0	0	0	0	2.3	0	0.0	0	0.0
3	0	0	0	0	0	0	0.0	0	0.0	0	0.0
4	0	0	0	1	0	0	0.0	0	0.0	0	0.0
5	0	0	0	0	0	0	2.1	0	1.4	0	0.0
6	0	0	0	0	0	0	0.0	0	0.0	0	0.0
V61SPW V62RBTR V63DWS V64BELL V65LWB V66CBW V67GGC V68PIL V69SKF V70RSL											
1	0	0	0.4	0.0	0	0	0	0	1.9	6.7	
2	0	0	0.0	0.0	0	0	0	0	0.0	0.0	
3	0	0	3.5	0.0	0	0	0	0	0.0	0.0	
4	0	0	5.5	0.0	4	0	0	0	0.0	0.8	
5	0	0	0.0	22.1	0	0	0	0	0.0	0.0	
6	0	0	0.0	0.0	0	0	0	0	0.0	0.0	
V71PDOV V72CRP V73JW V74BCHE V75RCR V76GBB V77RRP V78LLOR V79YTHE V80RF											
1	0	0	0	0	0	0.0	4.8	0	0	0.0	
2	0	0	0	0	0	0.0	0.0	0	0	0.0	
3	0	0	0	0	0	0.0	0.0	0	0	0.0	
4	0	0	0	0	0	0.0	0.0	0	0	0.0	
5	0	0	0	0	0	0.7	0.0	0	0	0.0	
6	0	0	0	0	0	0.0	0.0	0	0	1.4	
V81SHBC V82AZKF V83SFC V84YRTH V85ROSE V86BC00 V87LFC V88WG V89PC00 V90WTG											
1	0.0	0	0.0	0	0	0	0.0	0	1.9	0	
2	0.0	0	0.0	0	0	0	0.0	0	0.0	0	
3	1.4	0	0.0	0	0	0	1.8	0	0.0	0	
4	0.0	0	0.0	0	0	0	0.0	0	0.0	0	
5	0.7	0	0.0	0	0	0	0.0	0	0.0	0	
6	0.0	0	3.4	0	0	0	0.0	0	0.0	0	
V91NMIN V92NFB V93DB V94RBEE V95HBC V96DF V97PCL V98FLAME V99WWT V100WBWS											
1	0.2	0	0	0	0	0	9.1	0	0	0	

2	0.0	0	0	0	0	0	0.0	0	0	0
3	5.8	0	0	0	0	0	0.0	0	0	0
4	3.1	0	0	0	0	0	0.0	0	0	0
5	0.0	0	0	0	0	0	0.0	0	0	0
6	0.0	0	0	0	0	0	0.0	0	0	0
V101LCOR V102KING										
1	0	0								
2	0	0								
3	0	0								
4	0	0								
5	0	0								
6	0	0								

CODE

```
str(Birds)
```

OUTPUT

```
'data.frame':  37 obs. of  104 variables:
 $ SITE      : chr  "Reedy Lake" "Pearcedale" "Warneet" "Cranbourne" ...
 $ HABITAT   : chr  "Mixed" "Gippsland Ma" "Gippsland Ma" "Gippsland Ma" ...
 $ V16ST     : num  3.4 3.4 8.4 3 5.6 8.1 8.3 4.6 3.2 4.6 ...
 $ V2EYR     : num  0 9.2 3.8 5 5.6 4.1 7.1 5.3 5.2 1.2 ...
 $ V36F      : num  0 0 0.7 0 12.9 10.9 6.9 11.1 8.3 4.6 ...
 $ V4BTH     : num  0 0 2.8 5 12.2 24.5 29.1 28.2 18.2 6.5 ...
 $ V56WH     : num  0 0 0 2 9.5 5.6 4.2 3.9 3.8 2.3 ...
 $ V6WTTR    : num  0 0 0 0 2.1 6.7 2 6.5 4.2 5.2 ...
 $ V7WEHE    : num  0 0 10.7 3 7.9 9.4 7.1 2.6 2.8 0.6 ...
 $ V8WNHE    : num  11.9 11.5 12.3 10 28.6 6.7 27.4 10.9 9 3.6 ...
 $ V9SFW     : num  0.4 8.3 4.9 6.9 9.2 0 13.1 3.1 3.8 3.8 ...
 $ V10WBSW   : num  0 12.6 10.7 12 5 8.9 2.8 8.6 5.6 3 ...
 $ V11CR     : num  1.1 0 0 0 19.1 12.1 0 9.3 14.1 7.5 ...
 $ V12LK     : num  3.8 0.5 1.9 2 3.6 6.7 2.8 3.8 3.2 2.4 ...
 $ V13RWB    : num  9.7 11.6 16.6 11 5.7 2.7 2.4 0.6 0 0.6 ...
 $ V14AUR    : num  0 0 2.3 1.5 8.8 0 2.8 1.3 0 0 ...
 $ V15STTH   : num  0 0 2.8 0 7 16.8 13.9 10.2 12.2 11.3 ...
 $ V16LR     : num  4.8 3.7 5.5 11 1.6 3.4 0 0 0.6 5.8 ...
 $ V17WPHE   : num  27.3 27.6 27.5 20 0 0 16.7 0 0 0 ...
 $ V18YTH    : num  0 0 0 0 0 0 0 0 9.6 ...
 $ V19ER     : num  5.1 2.7 5.3 2.1 1.4 2.2 0 1.2 1.3 2.3 ...
 $ V20PCU    : num  0 0 0 0 0 0 0 0 2.8 2.9 ...
 $ V21ESP    : num  0 3.7 0 2 3.5 3.4 5.5 5.1 7.1 0.6 ...
 $ V22SCR    : num  0 0 0 0 0.7 0 0 0 0 3 ...
 $ V23RBFT   : num  0 1.1 0 5 0 0.7 0 0.7 1.9 0 ...
 $ V24BFCS   : num  0.6 1.1 1.5 1.4 2.7 2 3.6 0 0.6 1.2 ...
 $ V25WAG    : num  1.9 3.4 2.1 3.4 0 0 0 0 0 0 ...
 $ V26WWCH   : num  0 0 0 0 0 0 0 0 9.8 ...
 $ V27NHHE   : num  0 6.9 3 32 6.4 2.2 5.6 0 0 0 ...
 $ V28VS     : num  0 0 0 0 0 5.4 5.6 1.9 4.2 5.1 ...
 $ V29CST    : num  1.7 0.9 1.5 1.4 0 0 4.6 0 0 0 ...
 $ V30BTR    : num  12.5 0 0 0 0 0 0 0 0 0 ...
 $ V31AMAG   : num  8.6 0 0 0 0 0 0 0 0 0 ...
 $ V32SCC    : num  12.5 0 0 0 0 0 0 0 0 0 ...
 $ V33RWH    : num  0.6 2.3 1.4 0 7 6.8 7.3 9.1 4.5 6.3 ...
 $ V34WSW    : num  0 5.7 24.3 10 0 0 3.6 0 0 0 ...
 $ V35STP    : num  4.8 0 3.1 4 0 0 2.4 0 0 3.5 ...
 $ V36YFHE   : num  0 1.1 11.7 0 6.1 0 0 0 0 2.3 ...
 $ V37WHIP   : num  0 0 0 0 0 0 0 2 3.2 0 ...
 $ V38GAL    : num  4.8 0 0 2.8 0 0 0 0 0 0 ...
 $ V39FHE    : num  26.2 0 0 0 0 0 0 0 0 0 ...
 $ V40BRTH   : num  0 0 0 0 0 0 0 0 0 6 ...
 $ V41SPP    : num  0 1.1 4.6 0.8 5.4 3.4 0 2 2.6 4.2 ...
 $ V42SIL    : num  0 0 0 0 0 2.7 1.2 0 4.9 1.2 ...
```

```

$ V43GCU : num 0 0 0 0 0 1.4 0 2.6 1.3 0 ...
$ V44MUSK : num 13.1 0 0 0 0 0 0 0 0 0 ...
$ V45MGLK : num 1.7 0 0 1.4 0 0 0 0 0 0 ...
$ V46BHHE : num 1.1 0 0 0 0 0 0 0 0 0 ...
$ V47RFC : num 0 0 0 0 0 0 0 0 0 0 ...
$ V48YTBC : num 0 0 0 0 0 0 0 1.9 2.6 0 ...
$ V49LYRE : num 0 0 0 0 0 0 0 0 0.6 0 ...
$ V50CHE : num 0 0 0 0 0 0 0 0 0.6 0 ...
$ V51OWH : num 0 0 0 0 0 0 0 0 0 0 ...
$ V52TRM : num 15 0 0 0 0 0 0 0 0 0 ...
$ V53MB : num 0 0 0 1 0 0 3.6 0 0 1.2 ...
$ V54STHR : num 0 0 0 0 0 0 0 0 0 0 ...
$ V55LHE : num 0 0 0 0 0 0 0 0 0 0 ...
$ V56FTC : num 0 2.3 0 0 2.1 0 0 2.6 2.6 1.7 ...
$ V57PINK : num 0 0 0 0 0 0 0 0 0 0 ...
$ V58OB0 : num 0 0 0 0 1.4 0 0 0 0 1.2 ...
$ V59YR : num 0 0 0 0 0 0 0 0 0 0 ...
$ V60LFB : num 2.9 0 0 0 0 0 0 0 0 0 ...
$ V61SPW : num 0 0 0 0 0 0 0 0 0 0 ...
$ V62RBTR : num 0 0 0 0 0 0 0 0 0 0.6 ...
$ V63DWS : num 0.4 0 3.5 5.5 0 0 1.8 0 0 0 ...
$ V64BELL : num 0 0 0 0 22.1 0 0 0 0 0 ...
$ V65LWB : num 0 0 0 4 0 0 0 0 0 0 ...
$ V66CBW : num 0 0 0 0 0 0 0 0 0 0.6 ...
$ V67GGC : num 0 0 0 0 0 0 0 0 1.3 0 ...
$ V68PIL : num 0 0 0 0 0 0 0 0 1.3 0 ...
$ V69SKF : num 1.9 0 0 0 0 0 0 0 0 2.3 ...
$ V70RSL : num 6.7 0 0 0.8 0 0 0 0 0 0 ...
$ V71PD0V : num 0 0 0 0 0 0 0 0 0 0 ...
$ V72CRP : num 0 0 0 0 0 0 0 0 0 0 ...
$ V73JW : num 0 0 0 0 0 0 0 0 0 0 ...
$ V74BCHE : num 0 0 0 0 0 0 0 0 0 0 ...
$ V75RCR : num 0 0 0 0 0 0 0 0 0 0 ...
$ V76GBB : num 0 0 0 0 0.7 0 0 0 0.6 0 ...
$ V77RRP : num 4.8 0 0 0 0 0 0 0 0 0 ...
$ V78LLOR : num 0 0 0 0 0 0 0 0 0 0 ...
$ V79YTHE : num 0 0 0 0 0 0 0 0 0 0 ...
$ V80RF : num 0 0 0 0 0 1.4 0 1.9 3.2 0 ...
$ V81SHBC : num 0 0 1.4 0 0.7 0 0 2 1.3 0.6 ...
$ V82AZKF : num 0 0 0 0 0 0 0 0.7 0 0 ...
$ V83SFC : num 0 0 0 0 0 3.4 0 1.2 1.9 1.2 ...
$ V84YRTH : num 0 0 0 0 0 0 0 0 0 0 ...
$ V85ROSE : num 0 0 0 0 0 0 0 0.6 0 0 ...
$ V86BC00 : num 0 0 0 0 0 0 0 0 0.6 0 ...
$ V87LFC : num 0 0 1.8 0 0 0 2.4 0 0 0 ...
$ V88WG : num 0 0 0 0 0 0 0 0 0 0 ...
$ V89PC00 : num 1.9 0 0 0 0 0 0 0 0 0 ...
$ V90WTG : num 0 0 0 0 0 0 0 0 0 0 ...
$ V91NMIN : num 0.2 0 5.8 3.1 0 0 0 0 0 0 ...
$ V92NFB : num 0 0 0 0 0 0 0 0 0 0 ...
$ V93DB : num 0 0 0 0 0 0 0 0 0 0 ...
$ V94RBEE : num 0 0 0 0 0 0 0 0 0 0 ...
$ V95HBC : num 0 0 0 0 0 0 0 0 0 0 ...
$ V96DF : num 0 0 0 0 0 0 0 0 0 0 ...
$ V97PCL : num 9.1 0 0 0 0 0 0 0 0 0 ...

```

[list output truncated]

Check the habitat groups:

CODE

```
table(Birds$HABITAT)
```

OUTPUT

Box-Ironbark	Foothills	Wo	Gippsland	Ma	Mixed	Montane	Fore	River	Red	Gu
4		4		4	17		4		4	

Create the community matrix

For multivariate community analyses, we need only the species abundance columns.

The first two columns are site information, so the bird species matrix starts at column 3.

```
CODE
BirdsData <- Birds[, 3:ncol(Birds)]

head(BirdsData[, 1:8])
```

```
OUTPUT
```

	V1GST	V2EYR	V3GF	V4BTH	V5GWH	V6WTTR	V7WEHE	V8WNHE
1	3.4	0.0	0.0	0.0	0.0	0.0	0.0	11.9
2	3.4	9.2	0.0	0.0	0.0	0.0	0.0	11.5
3	8.4	3.8	0.7	2.8	0.0	0.0	10.7	12.3
4	3.0	5.0	0.0	5.0	2.0	0.0	3.0	10.0
5	5.6	5.6	12.9	12.2	9.5	2.1	7.9	28.6
6	8.1	4.1	10.9	24.5	5.6	6.7	9.4	6.7

```
CODE
dim(BirdsData)
```

```
OUTPUT
[1] 37 102
```

Each row is a site.

Each column is a bird species.

Transform the data

Community data often contain many zeros and some very abundant species. If we analyse raw abundance data, the ordination may be dominated by only the most abundant species.

A **fourth-root transformation** reduces the influence of very abundant species while still retaining quantitative abundance information.

CODE

```
TransBirdsData <- BirdsData^(1/4)
```

Bray-Curtis dissimilarity

Bray-Curtis dissimilarity is commonly used for ecological community data.

It compares sites based on differences in species composition and abundance.

CODE

```
Bird.dis <- vegdist(TransBirdsData, method = "bray")
```

Bird.dis

OUTPUT

	1	2	3	4	5	6	7
2	0.5811093						
3	0.5538884	0.3012592					
4	0.5168095	0.3048933	0.2656237				
5	0.7288140	0.4277846	0.3739545	0.4776578			
6	0.7346606	0.4854075	0.4626322	0.4971884	0.2904080		
7	0.6841188	0.4513940	0.3187727	0.3630946	0.3557227	0.3373155	
8	0.7836038	0.5372931	0.5115252	0.5490394	0.3083633	0.2240857	0.4019478
9	0.7828807	0.5558152	0.5564663	0.5824275	0.3465332	0.2390345	0.4506996
10	0.7018559	0.5557679	0.4865599	0.5552278	0.3271147	0.3391277	0.4324359
11	0.8063526	0.6167779	0.5001552	0.6020000	0.3385922	0.3581620	0.4327462
12	0.7266312	0.5499390	0.4809957	0.5191379	0.3633039	0.3201244	0.4158595
13	0.6924704	0.5196598	0.4507475	0.4978200	0.3222033	0.2564058	0.3464840
14	0.8510533	0.6203627	0.5630760	0.6157355	0.3812674	0.3493667	0.5065226
15	0.7959633	0.6222646	0.5527953	0.5849477	0.4094507	0.3742980	0.5031913
16	0.8073635	0.5931764	0.5088703	0.5757578	0.3731528	0.3838064	0.4486967
17	0.6218105	0.5123680	0.4292987	0.5145471	0.4254575	0.4521966	0.4541890
18	0.6288168	0.5141771	0.4502084	0.5041746	0.3400412	0.3792721	0.3382631
19	0.7169316	0.4737855	0.4365329	0.5063463	0.2919399	0.3493836	0.3982871
20	0.6882119	0.6015094	0.5146904	0.5463852	0.3506307	0.3781261	0.4076429
21	0.7207458	0.4628075	0.4030716	0.4833199	0.2213818	0.2272425	0.3184691
22	0.7130669	0.3963048	0.4435605	0.4157735	0.2887657	0.3190927	0.4548913
23	0.7554177	0.4648294	0.4673510	0.4573979	0.3576220	0.3357695	0.3699074
24	0.6578409	0.3261971	0.2845718	0.3033146	0.3974232	0.4653503	0.3550889
25	0.7564742	0.6557555	0.5304423	0.6354600	0.3886521	0.4254396	0.4272677
26	0.8274156	0.7185738	0.6055273	0.6441336	0.4908169	0.4240129	0.5252044
27	0.7135837	0.5642955	0.4683315	0.5301566	0.3160961	0.3563647	0.3996958
28	0.7518473	0.5009149	0.4421934	0.5193656	0.2893234	0.2855187	0.3778628
29	0.4531177	0.6256985	0.5994287	0.5463836	0.7257647	0.7478554	0.6379375
30	0.5565948	0.6965908	0.6312407	0.6125685	0.6996847	0.6982420	0.5969278
31	0.4446357	0.6983750	0.5853321	0.6002661	0.7165809	0.7143934	0.6339431
32	0.4199185	0.6335318	0.5783585	0.4944642	0.7458877	0.7876667	0.6852894
33	0.6319161	0.6323419	0.5275781	0.5963714	0.4590329	0.4614880	0.5310118
34	0.6294810	0.6351601	0.5177020	0.5992538	0.3705688	0.4405296	0.5484207
35	0.3936613	0.5282705	0.4333633	0.4680113	0.5600997	0.6449680	0.5806360
36	0.5940169	0.6182414	0.4841367	0.5814368	0.3857289	0.4355690	0.4637640
37	0.6700454	0.5603731	0.4565535	0.5501178	0.3244759	0.3876113	0.4089532

3
 4
 5
 6
 7
 8
 9 0.1951151
 10 0.3729687 0.3451244
 11 0.2717601 0.3015726 0.2454991
 12 0.2758298 0.2205846 0.3782381 0.2974830
 13 0.2223679 0.2226500 0.3022838 0.2391993 0.2064072
 14 0.2787909 0.2106467 0.4410868 0.3083956 0.2371092 0.3134122
 15 0.2957034 0.2363472 0.4193424 0.3057258 0.1709806 0.2850145 0.1613859
 16 0.2792215 0.2241250 0.4070212 0.2869727 0.2355982 0.2574916 0.2239040
 17 0.4841469 0.4539468 0.3891711 0.4753925 0.4742680 0.4427194 0.5274046
 18 0.3698541 0.4256049 0.4199316 0.3479013 0.3665011 0.3197675 0.4132451
 19 0.3448985 0.3535099 0.2735383 0.2536778 0.3190726 0.2979499 0.3937406
 20 0.4032488 0.4337080 0.2655589 0.2977125 0.3582272 0.2861492 0.4448466
 21 0.2674204 0.2833619 0.2760638 0.3092336 0.3071221 0.2366256 0.3122786
 22 0.3252211 0.3527112 0.4275477 0.4272384 0.4145930 0.3576349 0.3963489
 23 0.4007785 0.3676908 0.3715669 0.4481153 0.4011944 0.3888619 0.4561900
 24 0.5083378 0.5298122 0.5301139 0.5018120 0.4251041 0.3929244 0.5399769
 25 0.4648920 0.5136172 0.3287552 0.4046693 0.4807776 0.4143108 0.5661840
 26 0.3615145 0.3042304 0.4915418 0.4028192 0.2625627 0.3408723 0.2584830
 27 0.2781346 0.2495484 0.3170386 0.2401769 0.2479473 0.2026242 0.3126309
 28 0.2712258 0.2955475 0.3669449 0.3116880 0.2560819 0.2393910 0.3028019
 29 0.7759636 0.7209046 0.6281837 0.6958882 0.6936614 0.6787960 0.7493157
 30 0.7159321 0.7177126 0.5593490 0.6924704 0.7215431 0.6940181 0.7843145
 31 0.7356662 0.7161492 0.5865631 0.6970164 0.7090184 0.6692351 0.7842994
 32 0.8027009 0.8004714 0.7001694 0.7790406 0.7644425 0.7485170 0.8350025
 33 0.4901806 0.5099387 0.3462249 0.4307006 0.4683899 0.3941127 0.6080585
 34 0.4926852 0.4841842 0.3194907 0.4087302 0.4634862 0.4254745 0.5430026
 35 0.6613886 0.6645110 0.5425301 0.6348805 0.5855076 0.5589297 0.7292768
 36 0.4381436 0.4529427 0.2946949 0.3775072 0.3930000 0.3531846 0.5209080
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 15 16 17 18 19 20 21
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 19 0.3517541 0.3317676 0.4815143 0.3535642
 20 0.3778650 0.4184401 0.4768728 0.3427651 0.2776771
 21 0.3196154 0.3101225 0.4230529 0.3713238 0.2762805 0.3394350
 22 0.3948861 0.3740458 0.4790184 0.4235341 0.3677298 0.4639006 0.2553800
 23 0.4500571 0.4381147 0.4205407 0.4576355 0.4008931 0.4771469 0.2552954
 24 0.4904771 0.4784935 0.4323657 0.3680164 0.4560534 0.4966671 0.4067531
 25 0.5251346 0.5025636 0.5191401 0.4863091 0.3852502 0.3700371 0.3425710
 26 0.2428673 0.2709968 0.5601140 0.4732700 0.4319630 0.4850794 0.4318570
 27 0.2487586 0.1879095 0.4394022 0.3045709 0.2766347 0.3170154 0.2977140
 28 0.2851350 0.3187882 0.4377821 0.3762856 0.3792002 0.3476928 0.1653544
 29 0.7304746 0.7216215 0.4706448 0.5770535 0.6891767 0.6522839 0.7263503
 30 0.7711130 0.7399037 0.4099727 0.5995901 0.6740752 0.6443037 0.6454932

31	0.7697285	0.7376919	0.4553512	0.5903320	0.6651305	0.6405915	0.6875499
32	0.8007412	0.7928213	0.4998473	0.6263614	0.7416589	0.7330399	0.7707607
33	0.5159623	0.5118776	0.4950797	0.5026771	0.4331789	0.3777457	0.4190648
34	0.4991433	0.5215556	0.5160834	0.5069106	0.3507247	0.3386141	0.3738222
35	0.6424039	0.6436653	0.4773244	0.5105333	0.5668449	0.5273110	0.5959499
36	0.4753619	0.4808798	0.4595026	0.4644444	0.3737548	0.3067652	0.3430217
37	0.4091016	0.4238754	0.4456725	0.3813058	0.2548519	0.2078700	0.2918365
	22	23	24	25	26	27	28
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23	0.2931821						
24	0.4083533	0.3880791					
25	0.5574525	0.4645878	0.5039462				
26	0.5364468	0.5802964	0.5741870	0.6119036			
27	0.3812239	0.4063233	0.4310905	0.4381482	0.3262531		
28	0.2825661	0.3212464	0.4186415	0.4277676	0.4456717	0.3102918	
29	0.7579502	0.6825325	0.5779074	0.7216651	0.7289317	0.6850755	0.7333197
30	0.7669952	0.6707639	0.6821498	0.6050880	0.7661826	0.7008594	0.7055185
31	0.7772150	0.7069920	0.7177544	0.6333638	0.7337614	0.6958954	0.7424756
32	0.7820377	0.7662020	0.6293694	0.7544690	0.8168436	0.7574892	0.7990645
33	0.5408097	0.5194035	0.4946126	0.2631791	0.6205043	0.4609607	0.4206607
34	0.5204719	0.4994480	0.5324876	0.2478243	0.5855205	0.4575362	0.4133860
35	0.6017573	0.6591710	0.4996661	0.5393154	0.7531367	0.5558782	0.5947992
36	0.5108520	0.4842274	0.4955954	0.2930084	0.5419917	0.4133552	0.3836528
37	0.4343973	0.4248411	0.4837006	0.3358708	0.4954951	0.3376827	0.3357712
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31 0.3082678 0.2172576
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33 0.6752249 0.5995809 0.6008674 0.6838187
34 0.6768839 0.6314822 0.6503030 0.7069511 0.1945662
35 0.4733505 0.5019833 0.4990476 0.4264275 0.4308685 0.4676846
36 0.6242101 0.5747802 0.6046206 0.6708376 0.2282433 0.2260476 0.3915569
37 0.6887188 0.6069576 0.6060879 0.7142322 0.3269716 0.3210003 0.4916419
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37 0.2166400

```

Interpretation:

- values close to 0 = sites have similar bird communities
- values close to 1 = sites have very different bird communities

nMDS ordination

nMDS attempts to place sites in a low-dimensional plot so that distances among points reflect their Bray-Curtis dissimilarities.

Sites close together have similar bird assemblages.

Sites far apart have different bird assemblages.

CODE

```
set.seed(123)
nMDS_Bird <- metaMDS(TransBirdsData,
                     distance = "bray",
                     k = 2,
                     trymax = 100)
```

OUTPUT

```
Run 0 stress 0.1112209
Run 1 stress 0.1181205
Run 2 stress 0.1268505
Run 3 stress 0.1112209
... New best solution
... Procrustes: rmse 1.04629e-06 max resid 4.436971e-06
... Similar to previous best
Run 4 stress 0.1112209
... Procrustes: rmse 5.289056e-06 max resid 1.95247e-05
... Similar to previous best
Run 5 stress 0.1181205
Run 6 stress 0.1145319
Run 7 stress 0.111221
... Procrustes: rmse 5.574731e-05 max resid 0.0002576911
... Similar to previous best
Run 8 stress 0.1145319
Run 9 stress 0.111221
... Procrustes: rmse 9.96313e-05 max resid 0.0004700412
... Similar to previous best
Run 10 stress 0.1112209
... Procrustes: rmse 2.47593e-05 max resid 0.0001153228
... Similar to previous best
Run 11 stress 0.1298771
Run 12 stress 0.1112209
... New best solution
... Procrustes: rmse 2.937168e-06 max resid 1.411304e-05
... Similar to previous best
Run 13 stress 0.1181205
Run 14 stress 0.1112209
... New best solution
... Procrustes: rmse 2.35266e-06 max resid 1.158071e-05
... Similar to previous best
Run 15 stress 0.1112209
... Procrustes: rmse 1.305642e-05 max resid 6.065315e-05
... Similar to previous best
Run 16 stress 0.1145319
Run 17 stress 0.1181205
Run 18 stress 0.1181205
Run 19 stress 0.1268648
Run 20 stress 0.1181205
*** Best solution repeated 2 times
```

CODE

```
nMDS_Bird
```

OUTPUT

```
Call:
metaMDS(comm = TransBirdsData, distance = "bray", k = 2, trymax = 100)

global Multidimensional Scaling using monoMDS

Data:      TransBirdsData
Distance: bray

Dimensions: 2
Stress:     0.1112209
Stress type 1, weak ties
Best solution was repeated 2 times in 20 tries
The best solution was from try 14 (random start)
Scaling: centring, PC rotation, halfchange scaling
Species: expanded scores based on 'TransBirdsData'
```

Stress value

The stress value tells us how well the two-dimensional nMDS represents the multivariate dissimilarities.

As a guide:

- stress < 0.10 = good representation
- stress 0.10–0.20 = usable but interpret with care
- stress > 0.20 = poor representation

Base R nMDS plot

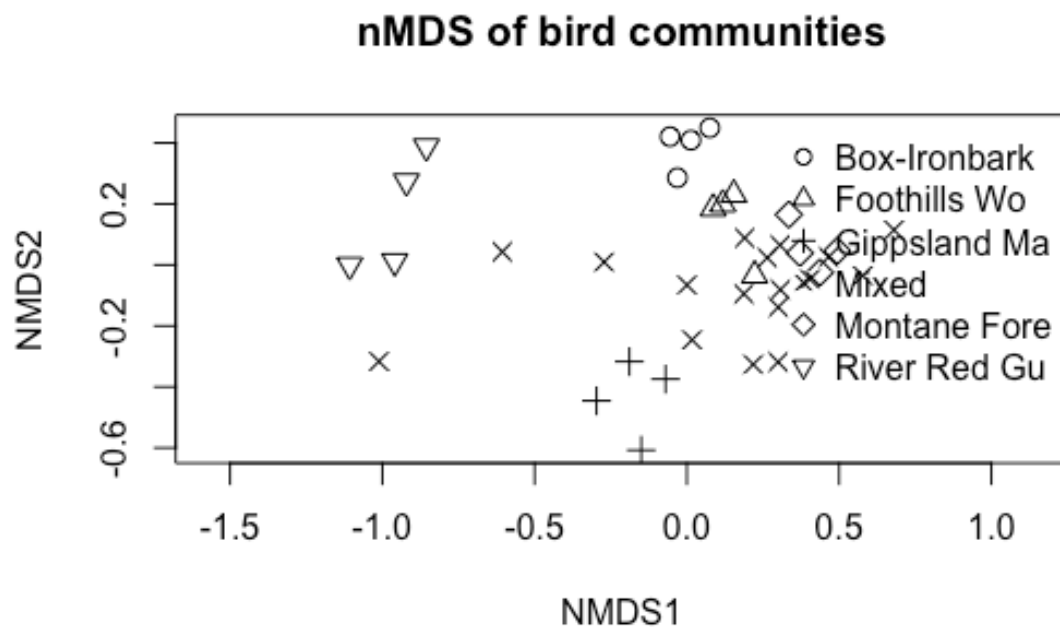
CODE

```
Bird_Factor <- factor(Birds$HABITAT)
pchs <- seq_along(levels(Bird_Factor))

plot(nMDS_Bird,
     display = "sites",
     type = "n",
     main = "nMDS of bird communities")

points(nMDS_Bird,
       display = "sites",
       pch = pchs[Bird_Factor],
       col = "black",
       cex = 1.2)
```

```
legend("topright",
      bty = "n",
      legend = levels(Bird_Factor),
      pch = pchs)
```



Interpretation

If sites from the same habitat type cluster together, this suggests that habitat type is associated with bird community composition.

If habitat groups strongly overlap, this suggests that bird communities are not clearly separated by habitat type.

ggplot nMDS plot

First extract the nMDS site scores.

CODE

```

nmDS_scores <- as.data.frame(scores(nMDS_Bird, display = "sites"))
nmDS_scores$SITE <- Birds$SITE
nmDS_scores$HABITAT <- Birds$HABITAT

head(nmDS_scores)

```

OUTPUT

	NMDS1	NMDS2	SITE	HABITAT
1	-1.0111806	-0.31642471	Reedy Lake	Mixed
2	-0.1497519	-0.60758304	Pearcedale	Gippsland Ma
3	-0.1887677	-0.31707278	Warneet	Gippsland Ma
4	-0.2965144	-0.44565339	Cranbourne	Gippsland Ma
5	0.1872838	-0.09485409	Lysterfield	Mixed
6	0.2999934	-0.13865712	Red Hill	Mixed

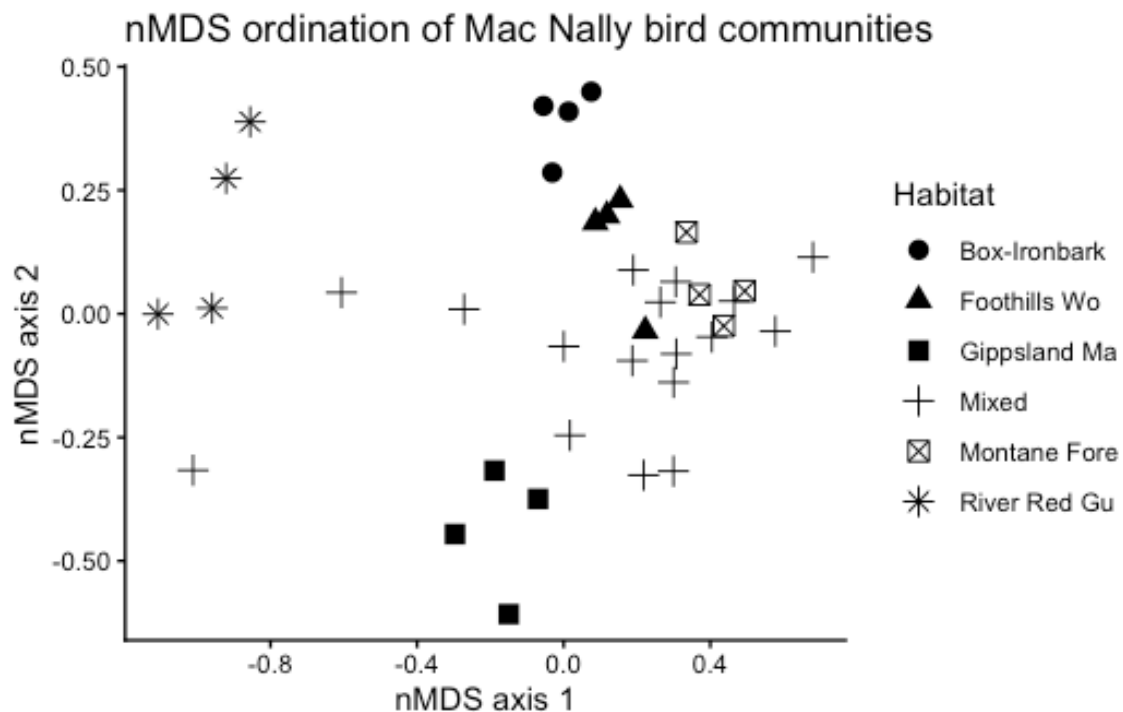
Now plot the ordination.

CODE

```

ggplot(nmDS_scores, aes(x = NMDS1, y = NMDS2, shape = HABITAT)) +
  geom_point(size = 3) +
  theme_classic() +
  labs(title = "nMDS ordination of Mac Nally bird communities",
       x = "nMDS axis 1",
       y = "nMDS axis 2",
       shape = "Habitat")

```

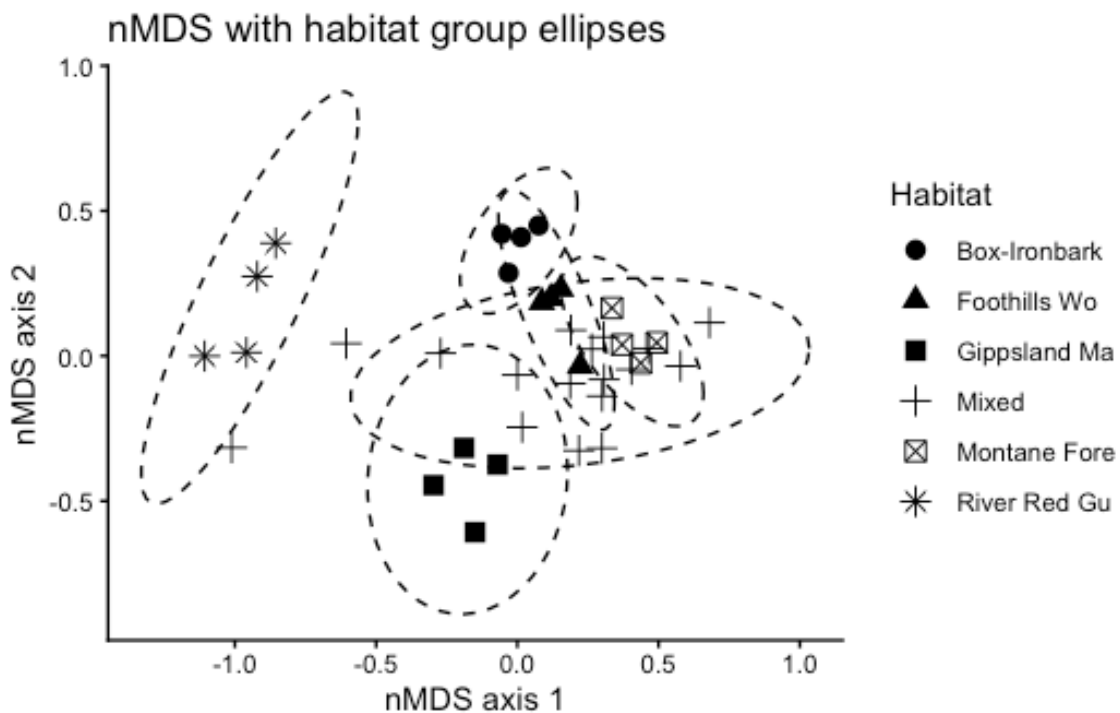


Add habitat ellipses

Ellipses help visualise whether habitat groups occupy different regions of ordination space.

CODE

```
ggplot(nmds_scores, aes(x = NMDS1, y = NMDS2, shape = HABITAT)) +  
  geom_point(size = 3) +  
  stat_ellipse(aes(group = HABITAT), type = "t", linetype = 2) +  
  theme_classic() +  
  labs(title = "nMDS with habitat group ellipses",  
       x = "nMDS axis 1",  
       y = "nMDS axis 2",  
       shape = "Habitat")
```



Note: ellipses are only a visual guide. Formal tests are below.

ANOSIM

ANOSIM tests whether differences among habitat groups are greater than differences within habitat groups.

CODE

```
Bird.anosim <- anosim(Bird.dis, Birds$HABITAT, permutations = 999)

summary(Bird.anosim)
```

OUTPUT

```
Call:
anosim(x = Bird.dis, grouping = Birds$HABITAT, permutations = 999)
Dissimilarity: bray
```

```
ANOSIM statistic R: 0.3936
Significance: 0.002
```

```
Permutation: free
Number of permutations: 999
```

```
Upper quantiles of permutations (null model):
 90%  95% 97.5% 99%
0.127 0.171 0.202 0.249
```

```
Dissimilarity ranks between and within classes:
```

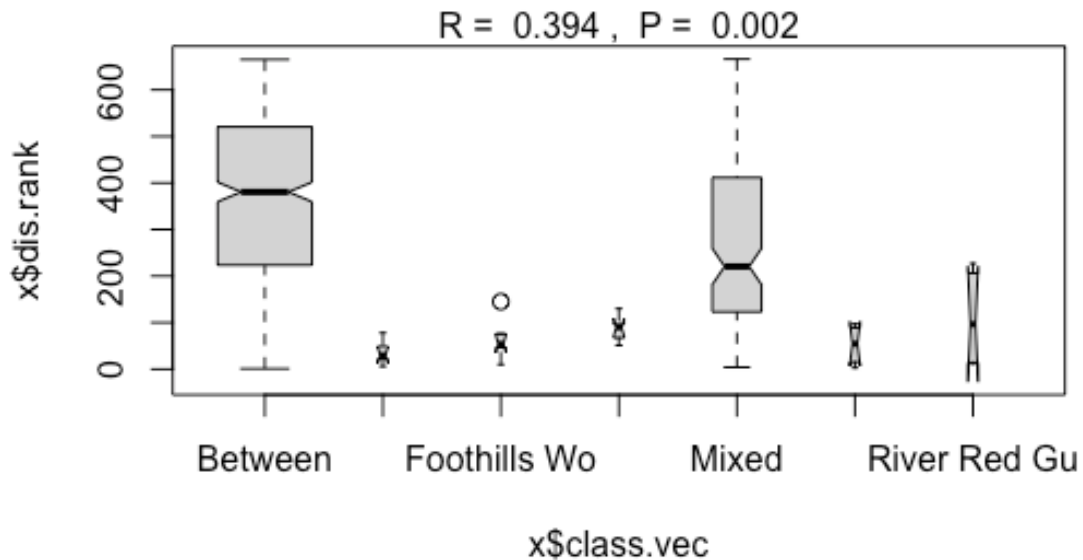
		0%	25%	50%	75%	100%	N
Between	1	224.25	380.5	520.25	665	500	
Box-Ironbark	6	22.50	29.5	45.50	78	6	
Foothills Wo	10	47.00	54.0	71.50	145	6	
Gippsland Ma	51	72.25	90.0	95.00	130	6	
Mixed	4	124.00	220.5	410.25	666	136	
Montane Fore	3	17.75	54.5	87.50	97	6	
River Red Gu	12	33.75	96.5	179.50	227	6	

CODE

```
plot(Bird.anosim)
```

OUTPUT

```
Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, : some
notches went outside hinges ('box'): maybe set notch=FALSE
```

Interpreting ANOSIM

The main statistic is **R**.

- R close to 0 = little separation among groups
- R close to 1 = strong separation among groups
- p-value = whether the observed separation is greater than expected by chance

PERMANOVA

PERMANOVA tests whether habitat type explains variation in the multivariate bird community data.

In vegan, PERMANOVA is run using `adonis2()`.

```
CODE
Bird.permanova <- adonis2(TransBirdsData ~ HABITAT,
  data = Birds,
  method = "bray",
```

```
permutations = 999)
```

```
Bird.permanova
```

```
OUTPUT
```

```
Permutation test for adonis under reduced model  
Permutation: free  
Number of permutations: 999
```

```
adonis2(formula = TransBirdsData ~ HABITAT, data = Birds, permutations = 999, method = "bray")
```

	Df	SumOfSqs	R2	F	Pr(>F)
Model	5	2.2749	0.50098	6.2245	0.001 ***
Residual	31	2.2660	0.49902		
Total	36	4.5409	1.00000		

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpreting PERMANOVA

Important columns:

- Df = degrees of freedom
- SumOfSqs = multivariate variation explained
- R2 = proportion of variation explained by habitat
- F = pseudo-F statistic
- Pr(>F) = permutation p-value

If the p-value is significant, bird communities differ among habitat types.

Check dispersion

PERMANOVA can be sensitive to differences in multivariate spread among groups. Therefore, we should check whether habitat groups differ in dispersion.

```
CODE
```

```
Bird.disp ← betadisper(Bird.dis, Birds$HABITAT)
```

```
anova(Bird.disp)
```

```
OUTPUT
```

```
Analysis of Variance Table
```

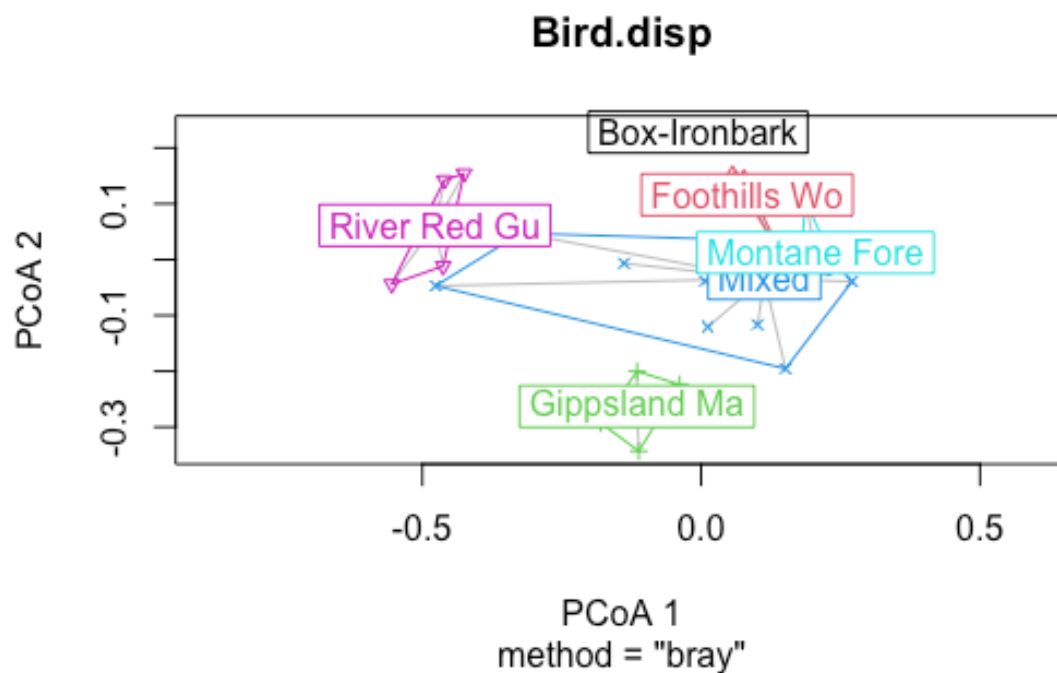
```
Response: Distances
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Groups	5	0.14637	0.0292742	3.0384	0.02403 *
Residuals	31	0.29868	0.0096347		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
CODE
```

```
plot(Bird.disp)
```



Interpretation

If this test is significant, habitat groups differ in their within-group variability.

That does not invalidate PERMANOVA, but it means the PERMANOVA result may reflect differences in spread as well as differences in group centroids.

Pairwise PERMANOVA comparisons

If the overall PERMANOVA is significant, we may want to know which habitat groups differ from each other.

The package `pairwiseAdonis` is useful, but it may not be installed on all computers. The installation code is shown but not run automatically.

```
CODE
# Install only once, if needed:
# install.packages("devtools")
# devtools::install_github("pmartinezarbizu/pairwiseAdonis/pairwiseAdonis")
```

Load the package if it is already installed:

```
CODE
library(pairwiseAdonis)

pairwise.adonis(TransBirdsData,
  Birds$HABITAT,
  sim.method = "bray",
  p.adjust.m = "bonferroni")
```

If this package is not available, you can still interpret the overall PERMANOVA and the nMDS plot.

SIMPER analysis

SIMPER stands for **Similarity Percentages**.

It identifies which species contribute most to differences between habitat groups.

```
CODE
Birdsim <- simper(TransBirdsData, Birds$HABITAT)

summary(Birdsim)
```

```
OUTPUT

Contrast: Mixed_Gippsland Ma
```

	average	sd	ratio	ava	avb	cumsum	p
V17WPHE	0.019779	0.011557	1.711400	0.602000	2.247700	0.040	0.003 **
V34WSW	0.018979	0.007112	2.668500	0.247500	1.856000	0.079	0.001 ***
V11CR	0.016850	0.007496	2.247800	1.443400	0.000000	0.114	0.001 ***
V6WTTT	0.015456	0.006359	2.430500	1.325800	0.000000	0.145	0.002 **
V15STTH	0.014940	0.009012	1.657800	1.475400	0.323400	0.176	0.008 **
V25WAG	0.013966	0.005802	2.407300	0.205100	1.376000	0.204	0.002 **
V27NHHE	0.012769	0.008435	1.513800	0.689200	1.646300	0.231	0.044 *
V36F	0.012483	0.008488	1.470700	1.569300	0.646500	0.256	0.001 ***
V4BTH	0.011472	0.009944	1.153700	1.946400	1.184200	0.280	0.129
V36YFHE	0.010708	0.008006	1.337500	1.121400	1.243700	0.302	0.086 .
V42SIL	0.010651	0.008608	1.237400	0.952800	0.420400	0.323	0.029 *
V13RWB	0.010585	0.007337	1.442700	1.080500	1.922100	0.345	0.156

V33RWH	0.010302	0.006859	1.501900	1.318700	0.579800	0.366	0.031	*
V56WH	0.010203	0.008334	1.224100	1.325200	0.674800	0.387	0.006	**
V16LR	0.009557	0.008123	1.176500	0.852900	1.184900	0.407	0.102	
V65LWB	0.009521	0.009391	1.013900	0.258700	0.741500	0.426	0.053	.
V21ESP	0.009484	0.007739	1.225400	1.236300	0.644000	0.445	0.130	
V23RBFT	0.009142	0.006803	1.343900	0.990600	0.937700	0.464	0.275	
V63DWS	0.008333	0.007483	1.113600	0.313100	0.724800	0.481	0.020	*
V35STP	0.008318	0.007536	1.103700	0.700100	0.993200	0.498	0.338	
V7WEHE	0.008195	0.007490	1.094100	1.252500	1.241500	0.515	0.162	
V91NMIN	0.008158	0.007918	1.030300	0.103300	0.719700	0.532	0.033	*
V19ER	0.007905	0.006671	1.184900	0.816000	1.308600	0.548	0.355	
V29CST	0.007715	0.005456	1.414100	0.548200	1.059100	0.564	0.079	.
V56FTC	0.007610	0.006779	1.122500	0.948600	0.615700	0.579	0.106	
V14AUR	0.007440	0.007024	1.059200	0.893600	0.962100	0.594	0.176	
V81SHBC	0.007124	0.006361	1.120000	0.641600	0.271900	0.609	0.107	
V28VS	0.007070	0.008926	0.792000	0.599300	0.000000	0.624	0.900	
V12LK	0.006733	0.005709	1.179300	1.372500	0.801000	0.637	0.009	**
V80RF	0.006689	0.007273	0.919800	0.568000	0.000000	0.651	0.251	
V50CHE	0.006645	0.008280	0.802400	0.580200	0.000000	0.664	0.318	
V46BHHE	0.006634	0.006911	0.959900	0.491200	0.307900	0.678	0.944	
V9SFW	0.006024	0.006604	0.912200	1.384400	1.734700	0.690	0.391	
V48YTBC	0.005842	0.007167	0.815200	0.511500	0.000000	0.702	0.378	
V41SPP	0.005728	0.005620	1.019200	1.058800	1.200500	0.714	0.702	
V8WNHE	0.005294	0.006131	0.863400	1.588600	1.843200	0.725	0.844	
V24BFCS	0.005165	0.005210	0.991300	0.751600	1.112500	0.736	0.545	
V37WHIP	0.005130	0.007087	0.723800	0.441900	0.000000	0.746	0.490	
V83SFC	0.005026	0.007077	0.710200	0.424600	0.000000	0.756	0.680	
V43GCU	0.004955	0.006908	0.717300	0.423100	0.000000	0.766	0.939	
V32SCC	0.004898	0.010846	0.451600	0.441300	0.000000	0.776	0.817	
V20PCU	0.004806	0.006607	0.727400	0.411800	0.000000	0.786	0.948	
V38GAL	0.004767	0.007107	0.670700	0.178700	0.323400	0.796	0.581	
V87LFC	0.004719	0.006574	0.717800	0.224900	0.289600	0.806	0.134	
V10WBSW	0.004538	0.005738	0.791000	1.426000	1.760400	0.815	1.000	
V2EYR	0.004523	0.005813	0.778100	1.285300	1.599500	0.824	0.895	
V53MB	0.004509	0.006393	0.705200	0.224200	0.250000	0.833	0.423	
V31AMAG	0.004040	0.007500	0.538600	0.370900	0.000000	0.842	0.771	
V55LHE	0.004024	0.006359	0.632900	0.354900	0.000000	0.850	0.482	
V45MGLK	0.003945	0.005906	0.668000	0.142600	0.271900	0.858	0.522	
V69SKF	0.003692	0.005819	0.634500	0.334500	0.000000	0.866	0.942	
V70RSL	0.003469	0.005750	0.603200	0.094600	0.236400	0.873	0.703	
V49LYRE	0.003229	0.005942	0.543500	0.268000	0.000000	0.879	0.646	
V85ROSE	0.002816	0.005168	0.545000	0.239700	0.000000	0.885	0.463	
V39FHE	0.002732	0.007692	0.355200	0.248500	0.000000	0.891	0.821	
V68PIL	0.002684	0.005897	0.455200	0.232900	0.000000	0.896	0.671	
V44MUSK	0.002563	0.007101	0.361000	0.236300	0.000000	0.901	0.730	
V98FLAME	0.002557	0.005598	0.456700	0.220500	0.000000	0.907	0.723	
V51OWH	0.002408	0.005315	0.453100	0.211500	0.000000	0.911	0.405	
V16ST	0.002323	0.001704	1.363300	1.466600	1.493300	0.916	0.309	
V22SCR	0.002318	0.005111	0.453500	0.210400	0.000000	0.921	0.999	
V52TRM	0.002247	0.006427	0.349700	0.202800	0.000000	0.926	0.714	
V47RFC	0.002229	0.004899	0.455000	0.214300	0.000000	0.930	0.891	
V30BTR	0.002214	0.006274	0.352900	0.200700	0.000000	0.935	0.943	
V580B0	0.002124	0.004652	0.456500	0.198800	0.000000	0.939	0.984	
V26WWCH	0.002007	0.005550	0.361500	0.180300	0.000000	0.943	1.000	
V77RRP	0.001900	0.005290	0.359200	0.174100	0.000000	0.947	0.223	
V76GBB	0.001874	0.004111	0.455800	0.166700	0.000000	0.951	0.548	
V18YTH	0.001726	0.004872	0.354300	0.171800	0.000000	0.954	1.000	
V86BC00	0.001595	0.004421	0.360800	0.142300	0.000000	0.958	0.573	
V67GGC	0.001558	0.004376	0.356000	0.135400	0.000000	0.961	0.787	
V64BELL	0.001553	0.006272	0.247500	0.127500	0.000000	0.964	0.409	
V62RBTR	0.001522	0.004260	0.357300	0.127400	0.000000	0.967	0.601	
V57PINK	0.001480	0.004113	0.359800	0.127300	0.000000	0.970	0.376	
V92NFB	0.001412	0.003904	0.361800	0.141400	0.000000	0.973	0.592	
V97PCL	0.001237	0.004997	0.247500	0.102200	0.000000	0.976	0.118	
V40BRTH	0.001151	0.004652	0.247400	0.088800	0.000000	0.978	0.989	

V78LLOR	0.001038	0.004191	0.247800	0.106900	0.000000	0.980	0.120
V60LFB	0.000929	0.003754	0.247500	0.076800	0.000000	0.982	0.961
V84YRTH	0.000851	0.003434	0.247700	0.082700	0.000000	0.984	0.621
V89PC00	0.000836	0.003378	0.247500	0.069100	0.000000	0.985	0.118
V54STHR	0.000766	0.003093	0.247700	0.073200	0.000000	0.987	0.847
V66CBW	0.000735	0.002968	0.247500	0.058800	0.000000	0.989	0.864
V79YTHE	0.000733	0.002957	0.247800	0.075400	0.000000	0.990	0.585
V96DF	0.000688	0.002777	0.247800	0.070800	0.000000	0.991	0.739
V82AZKF	0.000680	0.002747	0.247500	0.053800	0.000000	0.993	0.114
V100WBWS	0.000643	0.002594	0.247800	0.066200	0.000000	0.994	0.120
V93DB	0.000634	0.002557	0.247700	0.061600	0.000000	0.995	0.942
V102KING	0.000608	0.002454	0.247600	0.053800	0.000000	0.997	0.428
V90WTG	0.000572	0.002311	0.247700	0.055600	0.000000	0.998	0.958
V71PDOV	0.000540	0.002181	0.247800	0.055600	0.000000	0.999	0.786
V94RBEE	0.000523	0.002110	0.247800	0.053800	0.000000	1.000	0.981
V59YR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V61SPW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V72CRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V73JW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V74BCHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V75RCR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V88WG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V95HBC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V99WWT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V101LCOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed_Montane Fore

	average	sd	ratio	ava	avb	cumsum	p
V13RWB	0.009338	0.006981	1.337700	1.080500	0.662400	0.025	0.525
V43GCU	0.009042	0.005499	1.644400	0.423100	1.256100	0.049	0.011 *
V83SFC	0.008981	0.004949	1.814500	0.424600	1.220000	0.073	0.016 *
V20PCU	0.008970	0.005919	1.515400	0.411800	1.268100	0.097	0.015 *
V67GGC	0.008783	0.005683	1.545500	0.135400	0.926600	0.121	0.003 **
V16LR	0.008555	0.006520	1.312100	0.852900	1.023200	0.144	0.335
V46BHHE	0.008352	0.007086	1.178600	0.491200	0.828800	0.166	0.535
V33RWH	0.008286	0.006874	1.205500	1.318700	0.753700	0.188	0.224
V37WHIP	0.008217	0.006567	1.251200	0.441900	1.020400	0.211	0.026 *
V23RBFT	0.008056	0.006201	1.299200	0.990600	0.883700	0.232	0.712
V36YFHE	0.008026	0.007167	1.120000	1.121400	1.134500	0.254	0.883
V28VS	0.007929	0.007261	1.091900	0.599300	0.709200	0.275	0.616
V19ER	0.007909	0.006742	1.173100	0.816000	0.266900	0.296	0.390
V48YBTC	0.007759	0.006195	1.252500	0.511500	0.958400	0.317	0.038 *
V54STHR	0.007620	0.004661	1.634900	0.073200	0.758900	0.337	0.003 **
V68PIL	0.007514	0.004974	1.510900	0.232900	0.785200	0.358	0.007 **
V35STP	0.007322	0.006426	1.139400	0.700100	0.944200	0.377	0.787
V27NHHE	0.007298	0.008654	0.843300	0.689200	0.000000	0.397	0.692
V49LYRE	0.007170	0.005092	1.408100	0.268000	0.803100	0.416	0.009 **
V50CHE	0.006930	0.005950	1.164700	0.580200	0.509600	0.435	0.263
V80RF	0.006902	0.006031	1.144500	0.568000	0.913500	0.453	0.159
V42SIL	0.006868	0.005994	1.145900	0.952800	1.363600	0.472	0.960
V14AUR	0.006236	0.005868	1.062700	0.893600	0.919800	0.488	0.423
V55LHE	0.006136	0.005947	1.031900	0.354900	0.590500	0.505	0.094 .
V29CST	0.006120	0.006185	0.989500	0.548200	0.314400	0.521	0.873
V24BFCS	0.006092	0.005192	1.173200	0.751600	0.496700	0.537	0.229
V26WWCH	0.006052	0.009091	0.665800	0.180300	0.462400	0.554	0.978
V17WPHE	0.005965	0.009471	0.629800	0.602000	0.000000	0.570	0.991
V81SHBC	0.005961	0.005479	1.088100	0.641600	1.191300	0.586	0.867
V8WNHE	0.005525	0.005093	1.084800	1.588600	1.709100	0.601	0.795
V21ESP	0.005370	0.004726	1.136100	1.236300	1.502100	0.615	0.999
V15STTH	0.005290	0.006652	0.795300	1.475400	1.857500	0.629	0.956
V86BC00	0.005217	0.005078	1.027400	0.142300	0.481700	0.643	0.038 *
V11CR	0.004834	0.005601	0.863000	1.443400	1.710400	0.656	1.000

V41SPP	0.004678	0.005784	0.808700	1.058800	1.352200	0.669	0.889
V22SCR	0.004598	0.006404	0.718100	0.210400	0.317500	0.681	0.913
V85ROSE	0.004568	0.005803	0.787200	0.239700	0.331700	0.693	0.086
V9SFW	0.004450	0.004930	0.902600	1.384400	1.525000	0.705	0.794
V32SCC	0.004345	0.009627	0.451300	0.441300	0.000000	0.717	0.880
V4BTH	0.004124	0.005531	0.745600	1.946400	2.294000	0.728	0.996
V56FTC	0.004054	0.005016	0.808300	0.948600	1.231300	0.739	0.992
V6WTTR	0.003848	0.005024	0.765900	1.325800	1.567100	0.749	0.982
V62RBTR	0.003800	0.005744	0.661600	0.127400	0.281200	0.759	0.115
V7WEHE	0.003646	0.003640	1.001600	1.252500	1.224500	0.769	0.993
V31AMAG	0.003590	0.006655	0.539500	0.370900	0.000000	0.779	0.852
V51OWH	0.003548	0.004941	0.718000	0.211500	0.220000	0.788	0.225
V56WH	0.003405	0.004259	0.799600	1.325200	1.486100	0.797	0.962
V2EYR	0.003300	0.004546	0.725800	1.285300	1.440100	0.806	0.991
V69SKF	0.003277	0.005158	0.635200	0.334500	0.000000	0.815	0.983
V57PINK	0.003208	0.004800	0.668200	0.127300	0.250000	0.824	0.149
V76GBB	0.003203	0.004482	0.714700	0.166700	0.220000	0.832	0.173
V63DWS	0.003119	0.004987	0.625500	0.313100	0.000000	0.840	0.850
V10WBSW	0.002993	0.004868	0.614800	1.426000	1.622600	0.849	1.000
V36F	0.002942	0.004175	0.704800	1.569300	1.724600	0.856	0.996
V65LWB	0.002877	0.008006	0.359400	0.258700	0.000000	0.864	0.619
V102KING	0.002549	0.004128	0.617500	0.053800	0.236400	0.871	0.102
V39FHE	0.002424	0.006810	0.356000	0.248500	0.000000	0.877	0.888
V34WSW	0.002385	0.005233	0.455700	0.247500	0.000000	0.884	0.989
V53MB	0.002366	0.005345	0.442500	0.224200	0.000000	0.890	0.894
V44MUSK	0.002278	0.006306	0.361200	0.236300	0.000000	0.896	0.800
V98FLAME	0.002256	0.004937	0.457000	0.220500	0.000000	0.902	0.802
V87LFC	0.002212	0.004906	0.450900	0.224900	0.000000	0.908	0.700
V52TRM	0.001993	0.005682	0.350700	0.202800	0.000000	0.914	0.811
V47RFC	0.001992	0.004375	0.455300	0.214300	0.000000	0.919	0.936
V30BTR	0.001964	0.005551	0.353800	0.200700	0.000000	0.924	0.956
V25WAG	0.001953	0.004286	0.455700	0.205100	0.000000	0.929	0.998
V580B0	0.001891	0.004139	0.456800	0.198800	0.000000	0.934	0.989
V16ST	0.001754	0.001220	1.437400	1.466600	1.379200	0.939	0.857
V38GAL	0.001728	0.004792	0.360600	0.178700	0.000000	0.944	0.977
V77RRP	0.001687	0.004691	0.359700	0.174100	0.000000	0.948	0.414
V18YTH	0.001548	0.004368	0.354500	0.171800	0.000000	0.953	1.000
V45MGLK	0.001409	0.003896	0.361600	0.142600	0.000000	0.956	0.980
V64BELL	0.001363	0.005506	0.247500	0.127500	0.000000	0.960	0.542
V92NFB	0.001268	0.003504	0.361700	0.141400	0.000000	0.963	0.806
V12LK	0.001220	0.000891	1.368900	1.372500	1.324100	0.967	0.998
V97PCL	0.001087	0.004390	0.247500	0.102200	0.000000	0.970	0.501
V70RSL	0.001006	0.004066	0.247500	0.094600	0.000000	0.972	0.989
V40BRTH	0.001003	0.004051	0.247500	0.088800	0.000000	0.975	0.993
V91NMIN	0.000978	0.002735	0.357500	0.103300	0.000000	0.978	0.973
V78LLOR	0.000935	0.003773	0.247700	0.106900	0.000000	0.980	0.538
V60LFB	0.000816	0.003298	0.247500	0.076800	0.000000	0.982	0.955
V84YRTH	0.000761	0.003073	0.247700	0.082700	0.000000	0.984	0.800
V89PC00	0.000734	0.002967	0.247500	0.069100	0.000000	0.986	0.501
V79YTHE	0.000659	0.002662	0.247700	0.075400	0.000000	0.988	0.819
V66CBW	0.000643	0.002597	0.247500	0.058800	0.000000	0.990	0.927
V96DF	0.000619	0.002499	0.247700	0.070800	0.000000	0.991	0.828
V82AZKF	0.000594	0.002400	0.247500	0.053800	0.000000	0.993	0.515
V100WBWS	0.000579	0.002335	0.247700	0.066200	0.000000	0.995	0.538
V93DB	0.000567	0.002289	0.247700	0.061600	0.000000	0.996	0.970
V90WTG	0.000512	0.002068	0.247700	0.055600	0.000000	0.997	0.986
V71PD0V	0.000486	0.001964	0.247700	0.055600	0.000000	0.999	0.917
V94RBEE	0.000470	0.001899	0.247700	0.053800	0.000000	1.000	0.997
V59YR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V61SPW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V72CRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V73JW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V74BCHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V75RCR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V88WG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

V95HBC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V99WWT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V101LCOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed_Foothills Wo

	average	sd	ratio	ava	avb	cumsum	p
V40BRTH	0.014392	0.009230	1.559200	0.088800	1.423800	0.037	0.005 **
V22SCR	0.012437	0.004863	2.557800	0.210400	1.364500	0.068	0.002 **
V26WWCH	0.011907	0.007887	1.509700	0.180300	1.210300	0.099	0.020 *
V18YTH	0.011886	0.007877	1.508900	0.171800	1.199300	0.129	0.021 *
V10WBSW	0.009520	0.007570	1.257700	1.426000	0.773600	0.154	0.239
V23RBFT	0.009458	0.007442	1.270900	0.990600	0.243500	0.178	0.189
V28VS	0.009159	0.007107	1.288800	0.599300	1.088900	0.201	0.149
V53MB	0.008400	0.005483	1.532100	0.224200	0.851600	0.222	0.020 *
V46BHHE	0.008159	0.006034	1.352200	0.491200	0.931700	0.243	0.638
V16LR	0.008067	0.006758	1.193700	0.852900	1.479800	0.264	0.522
V19ER	0.007778	0.006808	1.142400	0.816000	0.654600	0.284	0.431
V69SKF	0.007708	0.006107	1.262200	0.334500	0.890600	0.304	0.050 *
V35STP	0.007702	0.006775	1.136900	0.700100	1.011200	0.323	0.677
V42SIL	0.007673	0.005328	1.440100	0.952800	0.761200	0.343	0.813
V27NHHE	0.007657	0.007842	0.976500	0.689200	0.256000	0.362	0.618
V50CHE	0.007225	0.007685	0.940200	0.580200	0.384600	0.381	0.256
V36YFHE	0.007150	0.005676	1.259600	1.121400	1.319700	0.399	0.967
V29CST	0.007090	0.006413	1.105700	0.548200	0.659900	0.417	0.272
V20PCU	0.007080	0.006583	1.075500	0.411800	0.673000	0.435	0.218
V9SFW	0.007048	0.006634	1.062400	1.384400	1.081900	0.453	0.219
V14AUR	0.006603	0.006170	1.070200	0.893600	0.987000	0.470	0.308
V43GCU	0.006573	0.006112	1.075300	0.423100	0.583000	0.487	0.494
V80RF	0.006348	0.006410	0.990200	0.568000	0.304500	0.503	0.399
V13RWB	0.006090	0.004647	1.310500	1.080500	1.170600	0.519	0.998
V17WPHE	0.006088	0.009642	0.631400	0.602000	0.000000	0.534	0.985
V64BELL	0.005858	0.009467	0.618800	0.127500	0.492000	0.549	0.134
V81SHBC	0.005798	0.005361	1.081400	0.641600	0.807100	0.564	0.928
V15STTH	0.005476	0.006465	0.847000	1.475400	1.788200	0.578	0.937
V83SFC	0.005382	0.006098	0.882700	0.424600	0.261700	0.592	0.581
V24BFCS	0.005280	0.004956	1.065500	0.751600	0.794400	0.605	0.495
V41SPP	0.005277	0.006727	0.784500	1.058800	1.526000	0.619	0.779
V48YTBC	0.005274	0.006448	0.817800	0.511500	0.000000	0.632	0.544
V8WNHE	0.005251	0.004942	1.062400	1.588600	1.582200	0.645	0.814
V11CR	0.004856	0.004784	1.015100	1.443400	1.484400	0.658	1.000
V21ESP	0.004753	0.004815	0.987100	1.236300	1.346100	0.670	1.000
V37WHIP	0.004623	0.006367	0.726100	0.441900	0.000000	0.682	0.665
V32SCC	0.004434	0.009801	0.452400	0.441300	0.000000	0.693	0.834
V63DWS	0.004331	0.005235	0.827300	0.313100	0.243500	0.704	0.552
V7WEHE	0.004274	0.004093	1.044200	1.252500	1.318200	0.715	0.951
V56FTC	0.004240	0.005159	0.821900	0.948600	1.244700	0.726	0.985
V33RWH	0.004130	0.005045	0.818600	1.318700	1.597500	0.736	0.985
V4BTH	0.004096	0.004957	0.826300	1.946400	1.975000	0.747	0.999
V96DF	0.003970	0.006435	0.617000	0.070800	0.349000	0.757	0.073
V580BO	0.003798	0.005307	0.715600	0.198800	0.261700	0.767	0.794
V6WTR	0.003785	0.005196	0.728400	1.325800	1.579900	0.776	0.970
V2EYR	0.003733	0.004281	0.872000	1.285300	1.316800	0.786	0.966
V31AMAG	0.003663	0.006775	0.540700	0.370900	0.000000	0.795	0.807
V55LHE	0.003636	0.005729	0.634700	0.354900	0.000000	0.804	0.610
V56WH	0.003634	0.003729	0.974400	1.325200	1.309900	0.814	0.928
V61SPW	0.003414	0.005979	0.571000	0.000000	0.326200	0.823	0.194
V3GF	0.003253	0.003748	0.868100	1.569300	1.569600	0.831	0.981
V62RBTR	0.003142	0.004683	0.670900	0.127400	0.220000	0.839	0.273
V65LWB	0.002942	0.008165	0.360300	0.258700	0.000000	0.846	0.622
V49LYRE	0.002899	0.005317	0.545200	0.268000	0.000000	0.854	0.707
V66CBW	0.002697	0.004351	0.619700	0.058800	0.220000	0.861	0.388
V85ROSE	0.002535	0.004640	0.546400	0.239700	0.000000	0.867	0.534

V39FHE	0.002474	0.006936	0.356600	0.248500	0.000000	0.874	0.860
V34WSW	0.002433	0.005328	0.456700	0.247500	0.000000	0.880	0.993
V16ST	0.002432	0.002103	1.156600	1.466600	1.577800	0.886	0.258
V68PIL	0.002421	0.005305	0.456500	0.232900	0.000000	0.892	0.713
V44MUSK	0.002324	0.006421	0.361900	0.236300	0.000000	0.898	0.772
V98FLAME	0.002304	0.005029	0.458100	0.220500	0.000000	0.904	0.783
V87LFC	0.002258	0.004995	0.452000	0.224900	0.000000	0.910	0.653
V510WH	0.002175	0.004792	0.453900	0.211500	0.000000	0.915	0.464
V52TRM	0.002033	0.005787	0.351300	0.202800	0.000000	0.920	0.750
V47RFC	0.002030	0.004451	0.456200	0.214300	0.000000	0.926	0.906
V30BTR	0.002004	0.005654	0.354500	0.200700	0.000000	0.931	0.939
V25WAG	0.001992	0.004362	0.456700	0.205100	0.000000	0.936	0.994
V38GAL	0.001763	0.004879	0.361300	0.178700	0.000000	0.940	0.973
V77RRP	0.001722	0.004777	0.360400	0.174100	0.000000	0.945	0.345
V76GBB	0.001693	0.003703	0.457300	0.166700	0.000000	0.949	0.639
V86BC00	0.001443	0.003991	0.361600	0.142300	0.000000	0.953	0.595
V45MGLK	0.001438	0.003967	0.362300	0.142600	0.000000	0.956	0.964
V12LK	0.001432	0.001053	1.360800	1.372500	1.350100	0.960	0.987
V67GGC	0.001406	0.003942	0.356500	0.135400	0.000000	0.964	0.849
V57PINK	0.001334	0.003700	0.360500	0.127300	0.000000	0.967	0.472
V92NFB	0.001291	0.003564	0.362400	0.141400	0.000000	0.970	0.761
V97PCL	0.001110	0.004475	0.248100	0.102200	0.000000	0.973	0.407
V70RSL	0.001028	0.004145	0.248100	0.094600	0.000000	0.976	0.989
V91NMIN	0.000997	0.002784	0.358300	0.103300	0.000000	0.978	0.965
V78LLOR	0.000952	0.003836	0.248100	0.106900	0.000000	0.981	0.423
V60LFB	0.000834	0.003362	0.248100	0.076800	0.000000	0.983	0.956
V84YRTH	0.000776	0.003127	0.248100	0.082700	0.000000	0.985	0.783
V89PC00	0.000750	0.003025	0.248100	0.069100	0.000000	0.987	0.407
V54STHR	0.000698	0.002812	0.248100	0.073200	0.000000	0.989	0.870
V79YTHE	0.000672	0.002706	0.248100	0.075400	0.000000	0.990	0.759
V82AZKF	0.000607	0.002448	0.248100	0.053800	0.000000	0.992	0.418
V100WBWS	0.000589	0.002375	0.248100	0.066200	0.000000	0.993	0.423
V93DB	0.000578	0.002329	0.248100	0.061600	0.000000	0.995	0.940
V102KING	0.000549	0.002214	0.248100	0.053800	0.000000	0.996	0.606
V90WTG	0.000522	0.002104	0.248100	0.055600	0.000000	0.998	0.974
V71PDOV	0.000495	0.001997	0.248100	0.055600	0.000000	0.999	0.872
V94RBEE	0.000479	0.001931	0.248100	0.053800	0.000000	1.000	0.982
V59YR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V72CRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V73JW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V74BCHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V75RCR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V88WG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V95HBC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V99WWT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V101LCOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed_Box-Ironbark

	average	sd	ratio	ava	avb	cumsum	p
V18YTH	0.018658	0.005852	3.188000	0.171800	1.909900	0.039	0.001 ***
V26WWCH	0.015902	0.005536	2.872000	0.180300	1.655500	0.072	0.001 ***
V10WBSW	0.015288	0.006045	2.529000	1.426000	0.000000	0.103	0.001 ***
V39FHE	0.014967	0.004566	3.278000	0.248500	1.532700	0.134	0.001 ***
V46BHHE	0.014098	0.007088	1.989000	0.491200	1.798900	0.163	0.002 **
V40BIRTH	0.013699	0.008353	1.640000	0.088800	1.379300	0.192	0.005 **
V21ESP	0.013293	0.006583	2.019000	1.236300	0.000000	0.219	0.001 ***
V22SCR	0.012484	0.005255	2.376000	0.210400	1.350800	0.245	0.001 ***
V44MUSK	0.011843	0.007846	1.509000	0.236300	1.180800	0.269	0.005 **
V75RCR	0.010748	0.003075	3.495000	0.000000	1.000900	0.292	0.001 ***
V580B0	0.009787	0.004514	2.168000	0.198800	1.059200	0.312	0.002 **
V23RBFT	0.009490	0.007770	1.221000	0.990600	0.320500	0.332	0.207
V13RWB	0.008878	0.006402	1.387000	1.080500	1.792200	0.350	0.677

V56FTC	0.008670	0.006269	1.383000	0.948600	0.281200	0.368	0.040	*
V28VS	0.008644	0.006295	1.373000	0.599300	0.971800	0.386	0.289	
V11CR	0.008437	0.007081	1.192000	1.443400	0.992900	0.403	0.769	
V36YFHE	0.008272	0.006141	1.347000	1.121400	1.383500	0.420	0.811	
V16LR	0.008079	0.006977	1.158000	0.852900	0.576500	0.437	0.543	
V35STP	0.007717	0.006654	1.160000	0.700100	0.656700	0.453	0.650	
V27NHHE	0.007497	0.008904	0.842000	0.689200	0.000000	0.468	0.670	
V19ER	0.007343	0.005945	1.235000	0.816000	0.846400	0.484	0.650	
V43GCU	0.007068	0.005570	1.269000	0.423100	0.818400	0.498	0.268	
V42SIL	0.006994	0.005548	1.261000	0.952800	1.255000	0.513	0.956	
V4BTH	0.006788	0.003994	1.699000	1.946400	1.520900	0.527	0.833	
V2EYR	0.006709	0.005999	1.118000	1.285300	0.874900	0.541	0.367	
V81SHBC	0.006489	0.005846	1.110000	0.641600	0.281200	0.554	0.398	
V66CBW	0.006361	0.006379	0.997000	0.058800	0.566600	0.567	0.008	**
V70RSL	0.006278	0.006199	1.013000	0.094600	0.557400	0.580	0.173	
V17WPHE	0.006117	0.009722	0.629000	0.602000	0.000000	0.593	0.981	
V80RF	0.006050	0.006578	0.920000	0.568000	0.000000	0.605	0.475	
V90WTG	0.006046	0.006464	0.935000	0.055600	0.542000	0.618	0.045	*
V50CHE	0.006025	0.007514	0.802000	0.580200	0.000000	0.630	0.497	
V29CST	0.005983	0.006099	0.981000	0.548200	0.256000	0.643	0.928	
V61SPW	0.005859	0.005982	0.980000	0.000000	0.593000	0.655	0.025	*
V14AUR	0.005388	0.005537	0.973000	0.893600	1.211600	0.666	0.706	
V48YTBC	0.005300	0.006506	0.815000	0.511500	0.000000	0.677	0.536	
V15STTH	0.005211	0.006022	0.865000	1.475400	1.627000	0.688	0.971	
V9SFW	0.005137	0.005014	1.024000	1.384400	1.472800	0.698	0.596	
V41SPP	0.004977	0.006330	0.786000	1.058800	1.440100	0.709	0.829	
V30BTR	0.004843	0.007338	0.660000	0.200700	0.382900	0.719	0.544	
V56WH	0.004733	0.003669	1.290000	1.325200	1.171900	0.729	0.707	
V3GF	0.004725	0.003539	1.335000	1.569300	1.330200	0.738	0.716	
V8WNHE	0.004707	0.005030	0.936000	1.588600	1.595400	0.748	0.915	
V37WHIP	0.004647	0.006424	0.723000	0.441900	0.000000	0.758	0.652	
V83SFC	0.004543	0.006404	0.710000	0.424600	0.000000	0.767	0.808	
V24BFCS	0.004457	0.004309	1.034000	0.751600	1.014400	0.776	0.785	
V32SCC	0.004454	0.009873	0.451000	0.441300	0.000000	0.785	0.843	
V20PCU	0.004351	0.005983	0.727000	0.411800	0.000000	0.794	0.982	
V79YTHE	0.004127	0.006650	0.621000	0.075400	0.396100	0.803	0.027	*
V63DWS	0.004112	0.005037	0.816000	0.313100	0.220000	0.811	0.647	
V7WEHE	0.004030	0.003983	1.012000	1.252500	1.296700	0.820	0.962	
V33RWH	0.003758	0.004830	0.778000	1.318700	1.512200	0.828	0.996	
V6WTR	0.003705	0.004996	0.741000	1.325800	1.525100	0.835	0.988	
V31AMAG	0.003680	0.006826	0.539000	0.370900	0.000000	0.843	0.798	
V55LHE	0.003654	0.005776	0.633000	0.354900	0.000000	0.850	0.576	
V69SKF	0.003359	0.005292	0.635000	0.334500	0.000000	0.857	0.968	
V76GBB	0.003230	0.004486	0.720000	0.166700	0.210200	0.864	0.176	
V65LWB	0.002960	0.008240	0.359000	0.258700	0.000000	0.870	0.620	
V49LYRE	0.002915	0.005365	0.543000	0.268000	0.000000	0.876	0.713	
V96DF	0.002818	0.004531	0.622000	0.070800	0.256000	0.882	0.241	
V12LK	0.002704	0.001574	1.718000	1.372500	1.122800	0.888	0.566	
V94RBEE	0.002692	0.004346	0.620000	0.053800	0.256000	0.893	0.406	
V85ROSE	0.002548	0.004679	0.545000	0.239700	0.000000	0.898	0.512	
V34WSW	0.002444	0.005368	0.455000	0.247500	0.000000	0.903	0.995	
V68PIL	0.002434	0.005347	0.455000	0.232900	0.000000	0.908	0.692	
V53MB	0.002430	0.005496	0.442000	0.224200	0.000000	0.914	0.901	
V16ST	0.002363	0.001590	1.487000	1.466600	1.425700	0.918	0.309	
V98FLAME	0.002315	0.005070	0.457000	0.220500	0.000000	0.923	0.806	
V87LFC	0.002268	0.005033	0.451000	0.224900	0.000000	0.928	0.680	
V51OWH	0.002186	0.004829	0.453000	0.211500	0.000000	0.932	0.443	
V88WG	0.002185	0.003827	0.571000	0.000000	0.210200	0.937	0.280	
V95HBC	0.002185	0.003827	0.571000	0.000000	0.210200	0.942	0.168	
V52TRM	0.002043	0.005832	0.350000	0.202800	0.000000	0.946	0.733	
V47RFC	0.002039	0.004481	0.455000	0.214300	0.000000	0.950	0.916	
V25WAG	0.002001	0.004394	0.455000	0.205100	0.000000	0.954	0.997	
V38GAL	0.001771	0.004914	0.360000	0.178700	0.000000	0.958	0.964	
V77RRP	0.001730	0.004812	0.359000	0.174100	0.000000	0.961	0.300	
V86BC00	0.001450	0.004021	0.361000	0.142300	0.000000	0.964	0.611	

V45MGLK	0.001444	0.003997	0.361000	0.142600	0.000000	0.967	0.967
V67GGC	0.001413	0.003972	0.356000	0.135400	0.000000	0.970	0.842
V64BELL	0.001400	0.005660	0.247000	0.127500	0.000000	0.973	0.518
V62RBTR	0.001374	0.003843	0.358000	0.127400	0.000000	0.976	0.672
V57PINK	0.001341	0.003729	0.359000	0.127300	0.000000	0.979	0.483
V92NFB	0.001296	0.003585	0.362000	0.141400	0.000000	0.981	0.716
V97PCL	0.001116	0.004512	0.247000	0.102200	0.000000	0.984	0.340
V91NMIN	0.001002	0.002802	0.357000	0.103300	0.000000	0.986	0.965
V78LLOR	0.000955	0.003858	0.248000	0.106900	0.000000	0.988	0.407
V60LFB	0.000839	0.003390	0.247000	0.076800	0.000000	0.990	0.949
V84YRTH	0.000779	0.003146	0.248000	0.082700	0.000000	0.991	0.757
V89PC00	0.000755	0.003050	0.247000	0.069100	0.000000	0.993	0.340
V54STHR	0.000701	0.002830	0.248000	0.073200	0.000000	0.994	0.839
V82AZKF	0.000611	0.002470	0.247000	0.053800	0.000000	0.995	0.367
V100WBWS	0.000591	0.002388	0.248000	0.066200	0.000000	0.997	0.407
V93DB	0.000580	0.002343	0.248000	0.061600	0.000000	0.998	0.971
V102KING	0.000552	0.002230	0.247000	0.053800	0.000000	0.999	0.570
V71PD0V	0.000497	0.002008	0.248000	0.055600	0.000000	1.000	0.864
V59YR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V72CRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V73JW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V74BCHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V99WWT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V101LCOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed_River Red Gu

	average	sd	ratio	ava	avb	cumsum	p
V4BTH	0.020445	0.006218	3.288000	1.946400	0.000000	0.030	0.001 ***
V32SCC	0.020280	0.007521	2.697000	0.441300	2.225000	0.060	0.002 **
V17WPHE	0.015917	0.009167	1.736000	0.602000	2.034900	0.084	0.018 *
V10WBSW	0.015087	0.005860	2.574000	1.426000	0.000000	0.106	0.001 ***
V11CR	0.015054	0.006597	2.282000	1.443400	0.000000	0.128	0.001 ***
V38GAL	0.014937	0.005417	2.758000	0.178700	1.582800	0.151	0.001 ***
V26WWCH	0.014881	0.004857	3.064000	0.180300	1.578000	0.173	0.003 **
V8WNHE	0.014481	0.007804	1.856000	1.588600	0.289600	0.194	0.003 **
V30BTR	0.014462	0.004730	3.057000	0.200700	1.520300	0.216	0.002 **
V2EYR	0.013450	0.005370	2.505000	1.285300	0.000000	0.236	0.001 ***
V7WEHE	0.013287	0.005547	2.395000	1.252500	0.000000	0.255	0.001 ***
V60LFB	0.013139	0.004117	3.192000	0.076800	1.312300	0.275	0.001 ***
V21ESP	0.013118	0.006422	2.043000	1.236300	0.000000	0.294	0.001 ***
V25WAG	0.012318	0.004959	2.484000	0.205100	1.365200	0.312	0.002 **
V31AMAG	0.011976	0.005536	2.163000	0.370900	1.393900	0.330	0.004 **
V45MGLK	0.011411	0.003990	2.860000	0.142600	1.216900	0.347	0.001 ***
V36YFHE	0.011370	0.007879	1.443000	1.121400	0.000000	0.364	0.034 *
V13RWB	0.011326	0.007442	1.522000	1.080500	0.000000	0.381	0.054 .
V47RFC	0.011238	0.004799	2.342000	0.214300	1.251400	0.397	0.002 **
V70RSL	0.011018	0.007241	1.522000	0.094600	1.085200	0.414	0.004 **
V41SPP	0.010892	0.006237	1.746000	1.058800	0.000000	0.430	0.006 **
V15STTH	0.010499	0.008202	1.280000	1.475400	0.745400	0.445	0.139
V35STP	0.010090	0.007467	1.351000	0.700100	1.638300	0.460	0.032 *
V73JW	0.009687	0.005914	1.638000	0.000000	0.934100	0.474	0.001 ***
V33RWH	0.009674	0.006163	1.570000	1.318700	0.532700	0.489	0.032 *
V42SIL	0.009667	0.007712	1.253000	0.952800	0.442300	0.503	0.118
V34WSW	0.009358	0.006570	1.424000	0.247500	0.961700	0.517	0.130
V52TRM	0.009124	0.005824	1.567000	0.202800	0.871300	0.531	0.005 **
V98FLAME	0.009079	0.006396	1.419000	0.220500	0.964700	0.544	0.006 **
V69SKF	0.008992	0.005367	1.675000	0.334500	1.156400	0.557	0.010 **
V18YTH	0.008930	0.008900	1.003000	0.171800	0.845000	0.571	0.294
V23RBFT	0.008751	0.006449	1.357000	0.990600	0.456500	0.584	0.393
V6WTR	0.008747	0.007272	1.203000	1.325800	0.736100	0.597	0.180
V56FTC	0.008722	0.005609	1.555000	0.948600	0.220000	0.609	0.035 *
V93DB	0.008385	0.005276	1.589000	0.061600	0.829800	0.622	0.002 **

V91NMIN	0.008172	0.007902	1.034000	0.103300	0.767400	0.634	0.026	*
V28VS	0.007982	0.007161	1.115000	0.599300	0.686000	0.646	0.629	
V56WH	0.007767	0.005012	1.550000	1.325200	0.765000	0.657	0.088	.
V94RBEE	0.007444	0.004665	1.596000	0.053800	0.732200	0.668	0.004	**
V27NHHE	0.007397	0.008745	0.846000	0.689200	0.000000	0.679	0.669	
V19ER	0.007375	0.006436	1.146000	0.816000	1.397300	0.690	0.663	
V16LR	0.006856	0.005658	1.212000	0.852900	1.252000	0.700	0.857	
V59YR	0.006854	0.006952	0.986000	0.000000	0.651800	0.711	0.002	**
V81SHBC	0.006705	0.005721	1.172000	0.641600	0.000000	0.720	0.258	
V36F	0.006585	0.003090	2.131000	1.569300	1.077300	0.730	0.285	
V29CST	0.006553	0.005132	1.277000	0.548200	0.776000	0.740	0.645	
V20PCU	0.006505	0.006055	1.074000	0.411800	0.602800	0.750	0.532	
V88WG	0.006448	0.006586	0.979000	0.000000	0.612900	0.759	0.003	**
V90WTG	0.006446	0.006479	0.995000	0.055600	0.612900	0.769	0.034	*
V99WWT	0.006439	0.006551	0.983000	0.000000	0.615300	0.778	0.008	**
V580B0	0.006407	0.006318	1.014000	0.198800	0.611700	0.788	0.100	.
V24BFCS	0.006242	0.005422	1.151000	0.751600	1.328700	0.797	0.201	
V9SFW	0.006159	0.003969	1.552000	1.384400	1.145500	0.806	0.352	
V80RF	0.005971	0.006465	0.924000	0.568000	0.000000	0.815	0.505	
V50CHE	0.005950	0.007394	0.805000	0.580200	0.000000	0.824	0.502	
V92NFB	0.005676	0.005583	1.017000	0.141400	0.513100	0.832	0.023	*
V72CRP	0.005564	0.005644	0.986000	0.000000	0.545600	0.840	0.009	**
V71PDOV	0.005528	0.005717	0.967000	0.055600	0.530500	0.849	0.015	*
V48YTBC	0.005233	0.006402	0.817000	0.511500	0.000000	0.856	0.571	
V14AUR	0.005141	0.005536	0.929000	0.893600	1.237700	0.864	0.755	
V12LK	0.005031	0.005700	0.883000	1.372500	0.910300	0.871	0.130	
V46BHHE	0.004971	0.006225	0.799000	0.491200	0.000000	0.879	0.998	
V37WHIP	0.004588	0.006321	0.726000	0.441900	0.000000	0.885	0.675	
V84YRTH	0.004512	0.007311	0.617000	0.082700	0.389600	0.892	0.027	*
V83SFC	0.004484	0.006299	0.712000	0.424600	0.000000	0.899	0.820	
V43GCU	0.004425	0.006145	0.720000	0.423100	0.000000	0.905	0.980	
V87LFC	0.004386	0.006114	0.717000	0.224900	0.314400	0.912	0.212	
V53MB	0.003834	0.005402	0.710000	0.224200	0.210200	0.917	0.646	
V101LCOR	0.003699	0.006477	0.571000	0.000000	0.370000	0.923	0.072	.
V55LHE	0.003608	0.005688	0.634000	0.354900	0.000000	0.928	0.593	
V63DWS	0.003160	0.005040	0.627000	0.313100	0.000000	0.933	0.851	
V65LWB	0.002918	0.008100	0.360000	0.258700	0.000000	0.937	0.596	
V49LYRE	0.002876	0.005277	0.545000	0.268000	0.000000	0.942	0.712	
V74BCHE	0.002855	0.005003	0.571000	0.000000	0.261700	0.946	0.065	.
V85ROSE	0.002515	0.004606	0.546000	0.239700	0.000000	0.950	0.550	
V39FHE	0.002455	0.006886	0.357000	0.248500	0.000000	0.953	0.897	
V68PIL	0.002403	0.005266	0.456000	0.232900	0.000000	0.957	0.709	
V44MUSK	0.002307	0.006375	0.362000	0.236300	0.000000	0.960	0.800	
V95HBC	0.002170	0.003801	0.571000	0.000000	0.198800	0.963	0.175	
V510WH	0.002158	0.004757	0.454000	0.211500	0.000000	0.967	0.454	
V22SCR	0.002084	0.004592	0.454000	0.210400	0.000000	0.970	1.000	
V16ST	0.002006	0.001368	1.466000	1.466600	1.343300	0.973	0.616	
V77RRP	0.001709	0.004743	0.360000	0.174100	0.000000	0.975	0.380	
V76GBB	0.001680	0.003676	0.457000	0.166700	0.000000	0.978	0.677	
V86BC00	0.001432	0.003963	0.361000	0.142300	0.000000	0.980	0.620	
V67GGC	0.001395	0.003914	0.356000	0.135400	0.000000	0.982	0.860	
V64BELL	0.001382	0.005570	0.248000	0.127500	0.000000	0.984	0.559	
V62RBTR	0.001356	0.003781	0.359000	0.127400	0.000000	0.986	0.709	
V57PINK	0.001324	0.003673	0.360000	0.127300	0.000000	0.988	0.485	
V97PCL	0.001101	0.004441	0.248000	0.102200	0.000000	0.990	0.443	
V40BRTH	0.001017	0.004100	0.248000	0.088800	0.000000	0.991	0.999	
V78LLOR	0.000946	0.003812	0.248000	0.106900	0.000000	0.992	0.474	
V89PC00	0.000744	0.003002	0.248000	0.069100	0.000000	0.994	0.443	
V54STHR	0.000693	0.002793	0.248000	0.073200	0.000000	0.995	0.859	
V79YTHE	0.000667	0.002689	0.248000	0.075400	0.000000	0.996	0.803	
V66CBW	0.000652	0.002628	0.248000	0.058800	0.000000	0.997	0.916	
V96DF	0.000626	0.002525	0.248000	0.070800	0.000000	0.997	0.847	
V82AZKF	0.000602	0.002428	0.248000	0.053800	0.000000	0.998	0.475	
V100WBWS	0.000585	0.002359	0.248000	0.066200	0.000000	0.999	0.474	
V102KING	0.000545	0.002198	0.248000	0.053800	0.000000	1.000	0.603	

V61SPW 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V75RCR 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma_Montane Fore

	average	sd	ratio	ava	avb	cumsum	p
V17WPHE	0.025511	0.002976	8.572000	2.247700	0.000000	0.047	0.002 **
V34WSW	0.020923	0.002668	7.841000	1.856000	0.000000	0.086	0.001 ***
V11CR	0.019365	0.002542	7.617000	0.000000	1.710400	0.121	0.002 **
V27NHHE	0.018629	0.005197	3.584000	1.646300	0.000000	0.156	0.002 **
V6WTTR	0.017764	0.002227	7.977000	0.000000	1.567100	0.189	0.002 **
V15STTH	0.017530	0.007233	2.423000	0.323400	1.857500	0.221	0.005 **
V25WAG	0.015594	0.002143	7.276000	1.376000	0.000000	0.250	0.001 ***
V13RWB	0.014804	0.009495	1.559000	1.922100	0.662400	0.277	0.011 *
V20PCU	0.014406	0.001861	7.739000	0.000000	1.268100	0.304	0.003 **
V43GCU	0.014108	0.001624	8.689000	0.000000	1.256100	0.330	0.002 **
V83SFC	0.013763	0.001275	10.794000	0.000000	1.220000	0.355	0.002 **
V4BTH	0.012989	0.009916	1.310000	1.184200	2.294000	0.379	0.102
V42SIL	0.012586	0.005694	2.210000	0.420400	1.363600	0.402	0.034 *
V36F	0.012521	0.008426	1.486000	0.646500	1.724600	0.425	0.007 **
V19ER	0.011718	0.005658	2.071000	1.308600	0.266900	0.447	0.029 *
V37WHIP	0.011155	0.006729	1.658000	0.000000	1.020400	0.468	0.003 **
V48YTBC	0.010991	0.006611	1.663000	0.000000	0.958400	0.488	0.006 **
V81SHBC	0.010480	0.005715	1.834000	0.271900	1.191300	0.507	0.012 *
V21ESP	0.010274	0.006547	1.569000	0.644000	1.502100	0.526	0.142
V67GGC	0.010101	0.006264	1.613000	0.000000	0.926600	0.545	0.002 **
V80RF	0.010024	0.006191	1.619000	0.000000	0.913500	0.563	0.026 *
V56WH	0.009874	0.007909	1.248000	0.674800	1.486100	0.582	0.028 *
V29CST	0.009751	0.004639	2.102000	1.059100	0.314400	0.599	0.047 *
V36YFHE	0.009681	0.007030	1.377000	1.243700	1.134500	0.617	0.410
V16LR	0.009375	0.007159	1.310000	1.184900	1.023200	0.635	0.243
V46BHHE	0.008981	0.008370	1.073000	0.307900	0.828800	0.651	0.389
V49LYRE	0.008738	0.005375	1.626000	0.000000	0.803100	0.667	0.013 *
V33RWH	0.008651	0.007180	1.205000	0.579800	0.753700	0.683	0.264
V68PIL	0.008604	0.005317	1.618000	0.000000	0.785200	0.699	0.012 *
V54STHR	0.008499	0.005150	1.650000	0.000000	0.758900	0.715	0.001 ***
V65LWB	0.008071	0.008400	0.961000	0.741500	0.000000	0.730	0.153
V63DWS	0.007899	0.008217	0.961000	0.724800	0.000000	0.744	0.063 .
V91NMIN	0.007851	0.008209	0.956000	0.719700	0.000000	0.759	0.081 .
V28VS	0.007829	0.008154	0.960000	0.000000	0.709200	0.773	0.611
V7WEHE	0.007421	0.005457	1.360000	1.241500	1.224500	0.787	0.378
V24BFCS	0.007277	0.005961	1.221000	1.112500	0.496700	0.800	0.129
V56FTC	0.007098	0.006534	1.086000	0.615700	1.231300	0.813	0.428
V35STP	0.006718	0.007148	0.940000	0.993200	0.944200	0.826	0.805
V23RBFT	0.006640	0.006306	1.053000	0.937700	0.883700	0.838	0.889
V14AUR	0.006610	0.006785	0.974000	0.962100	0.919800	0.850	0.421
V55LHE	0.006240	0.006468	0.965000	0.000000	0.590500	0.862	0.186
V12LK	0.005918	0.005486	1.079000	0.801000	1.324100	0.873	0.071 .
V26WWCH	0.005801	0.010413	0.557000	0.000000	0.462400	0.883	0.892
V50CHE	0.005678	0.005941	0.956000	0.000000	0.509600	0.894	0.587
V86BC00	0.005379	0.005588	0.962000	0.000000	0.481700	0.904	0.081 .
V22SCR	0.003983	0.007149	0.557000	0.000000	0.317500	0.911	0.851
V8WNHE	0.003758	0.001583	2.374000	1.843200	1.709100	0.918	0.877
V85ROSE	0.003564	0.006391	0.558000	0.000000	0.331700	0.925	0.300
V62RBTR	0.003528	0.006332	0.557000	0.000000	0.281200	0.931	0.212
V38GAL	0.003497	0.006277	0.557000	0.323400	0.000000	0.938	0.801
V87LFC	0.003183	0.005712	0.557000	0.289600	0.000000	0.944	0.573
V45MGLK	0.002941	0.005278	0.557000	0.271900	0.000000	0.949	0.813
V41SPP	0.002919	0.001917	1.523000	1.200500	1.352200	0.954	0.890
V9SFW	0.002819	0.002623	1.075000	1.734700	1.525000	0.960	0.917
V53MB	0.002704	0.004852	0.557000	0.250000	0.000000	0.965	0.808
V57PINK	0.002686	0.004816	0.558000	0.000000	0.250000	0.970	0.270
V76GBB	0.002568	0.004606	0.557000	0.000000	0.220000	0.974	0.381

V70RSL	0.002557	0.004589	0.557000	0.236400	0.000000	0.979	0.799
V2EYR	0.002508	0.001941	1.292000	1.599500	1.440100	0.984	0.914
V102KING	0.002459	0.004408	0.558000	0.000000	0.236400	0.988	0.069
V510WH	0.002364	0.004239	0.558000	0.000000	0.220000	0.993	0.535
V10WBSW	0.002324	0.001192	1.950000	1.760400	1.622600	0.997	0.968
V16ST	0.001754	0.001373	1.278000	1.493300	1.379200	1.000	0.741
V18YTH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V30BTR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V31AMAG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V32SCC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V39FHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V40BRTH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V44MUSK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V47RFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V52TRM	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V580B0	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V59YR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V60LFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V61SPW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V64BELL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V66CBW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V69SKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V71PD0V	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V72CRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V73JW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V74BCHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V75RCR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V77RRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V78LLOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V79YTHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V82AZKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V84YRTH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V88WG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V89PC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V90WTG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V92NFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V93DB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V94RBEE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V95HBC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V96DF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V97PCL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V98FLAME	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V99WWT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V100WBWS	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V101LCOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma_Foothills Wo

	average	sd	ratio	ava	avb	cumsum	p
V17WPHE	0.026097	0.002383	10.951000	2.247700	0.000000	0.052	0.001 ***
V34WSW	0.021402	0.002230	9.597000	1.856000	0.000000	0.094	0.001 ***
V6WTTR	0.018324	0.001427	12.839000	0.000000	1.579900	0.130	0.001 ***
V11CR	0.017226	0.001813	9.499000	0.000000	1.484400	0.164	0.003 **
V15STTH	0.017112	0.007416	2.307000	0.323400	1.788200	0.198	0.003 **
V40BRTH	0.016327	0.010384	1.572000	0.000000	1.423800	0.230	0.003 **
V27NHHE	0.015988	0.007215	2.216000	1.646300	0.256000	0.262	0.012 *
V25WAG	0.015952	0.001857	8.589000	1.376000	0.000000	0.293	0.001 ***
V22SCR	0.015801	0.001670	9.460000	0.000000	1.364500	0.325	0.001 ***
V26WWCH	0.013884	0.008667	1.602000	0.000000	1.210300	0.352	0.018 *
V18YTH	0.013759	0.008550	1.609000	0.000000	1.199300	0.379	0.021 *
V28VS	0.012735	0.007685	1.657000	0.000000	1.088900	0.404	0.016 *
V10WBSW	0.011739	0.009043	1.298000	1.760400	0.773600	0.428	0.062 .
V36F	0.011589	0.007893	1.468000	0.646500	1.569600	0.451	0.010 **

V33RWH	0.011462	0.006709	1.709000	0.579800	1.597500	0.473	0.016	*
V69SKF	0.010219	0.006162	1.658000	0.000000	0.890600	0.494	0.008	**
V4BTH	0.010179	0.009715	1.048000	1.184200	1.975000	0.514	0.349	
V23RBT	0.009465	0.006668	1.419000	0.937700	0.243500	0.532	0.315	
V46BHHE	0.009439	0.007230	1.305000	0.307900	0.931700	0.551	0.278	
V42SIL	0.009295	0.005353	1.736000	0.420400	0.761200	0.569	0.289	
V21ESP	0.009082	0.006693	1.357000	0.644000	1.346100	0.587	0.294	
V5GWH	0.008800	0.007323	1.202000	0.674800	1.309900	0.605	0.082	.
V13RWB	0.008699	0.002320	3.750000	1.922100	1.170600	0.622	0.646	
V53MB	0.008525	0.006365	1.339000	0.250000	0.851600	0.639	0.026	*
V36YFHE	0.008313	0.004873	1.706000	1.243700	1.319700	0.655	0.696	
V65LWB	0.008250	0.008548	0.965000	0.741500	0.000000	0.671	0.149	
V63DWS	0.008179	0.007487	1.092000	0.724800	0.243500	0.688	0.041	*
V81SHBC	0.008171	0.006178	1.323000	0.271900	0.807100	0.704	0.076	.
V19ER	0.008140	0.007125	1.143000	1.308600	0.654600	0.720	0.354	
V91NMIN	0.008026	0.008354	0.961000	0.719700	0.000000	0.736	0.062	.
V7WEHE	0.007740	0.006507	1.189000	1.241500	1.318200	0.751	0.319	
V29CST	0.007736	0.004933	1.568000	1.059100	0.659900	0.766	0.202	
V20PCU	0.007716	0.008005	0.964000	0.000000	0.673000	0.782	0.170	
V9SFW	0.007643	0.007705	0.992000	1.734700	1.081900	0.797	0.205	
V56FTC	0.007448	0.006647	1.120000	0.615700	1.244700	0.811	0.262	
V14AUR	0.006977	0.007154	0.975000	0.962100	0.987000	0.825	0.356	
V35STP	0.006914	0.007551	0.916000	0.993200	1.011200	0.839	0.760	
V43GCU	0.006636	0.007051	0.941000	0.000000	0.583000	0.852	0.494	
V12LK	0.006350	0.005685	1.117000	0.801000	1.350100	0.865	0.044	*
V16LR	0.006246	0.006500	0.961000	1.184900	1.479800	0.877	0.839	
V64BELL	0.005568	0.009988	0.557000	0.000000	0.492000	0.888	0.215	
V50CHE	0.004608	0.008269	0.557000	0.000000	0.384600	0.897	0.689	
V96DF	0.004006	0.007186	0.557000	0.000000	0.349000	0.905	0.103	
V41SPP	0.003877	0.002906	1.334000	1.200500	1.526000	0.913	0.886	
V24BFCS	0.003777	0.005287	0.714000	1.112500	0.794400	0.920	0.836	
V61SPW	0.003744	0.006716	0.557000	0.000000	0.326200	0.927	0.117	
V80RF	0.003648	0.006547	0.557000	0.000000	0.304500	0.935	0.850	
V2EYR	0.003593	0.002576	1.394000	1.599500	1.316800	0.942	0.853	
V38GAL	0.003574	0.006396	0.559000	0.323400	0.000000	0.949	0.765	
V87LFC	0.003254	0.005822	0.559000	0.289600	0.000000	0.955	0.541	
V8WNHE	0.003139	0.002678	1.172000	1.843200	1.582200	0.962	0.917	
V580B0	0.003041	0.005456	0.557000	0.000000	0.261700	0.968	0.800	
V83SFC	0.003041	0.005456	0.557000	0.000000	0.261700	0.974	0.867	
V45MGLK	0.003006	0.005378	0.559000	0.271900	0.000000	0.980	0.756	
V16ST	0.002663	0.001997	1.334000	1.493300	1.577800	0.985	0.199	
V70RSL	0.002613	0.004676	0.559000	0.236400	0.000000	0.990	0.807	
V62RBT	0.002557	0.004588	0.557000	0.000000	0.220000	0.995	0.452	
V66CBW	0.002557	0.004588	0.557000	0.000000	0.220000	1.000	0.528	
V30BTR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V31AMAG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V32SCC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V37WHIP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V39FHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V44MUSK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V47RFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V48YTB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V49LYRE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V51OWH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V52TRM	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V54STHR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V55LHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V57PINK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V59YR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V60LFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V67GGC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V68PIL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V71PD0V	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V72CRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V73JW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	

V74BCHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V75RCR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V76GBB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V77RRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V78LLOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V79YTHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V82AZKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V84YRTH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V85ROSE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V86BC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V88WG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V89PC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V90WTG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V92NFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V93DB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V94RBEE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V95HBC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V97PCL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V98FLAME	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V99WWT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V100WBWS	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V101LCOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V102KING	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma_Box-Ironbark

	average	sd	ratio	ava	avb	cumsum	p
V17WPHE	0.026261	0.003224	8.146000	2.247700	0.000000	0.046	0.001 ***
V18YTH	0.022239	0.002654	8.378000	0.000000	1.909900	0.086	0.001 ***
V34WSW	0.021534	0.002823	7.627000	1.856000	0.000000	0.124	0.001 ***
V10WBSW	0.020600	0.003335	6.178000	1.760400	0.000000	0.160	0.001 ***
V26WWCH	0.019377	0.002919	6.638000	0.000000	1.655500	0.195	0.001 ***
V27NHHE	0.019175	0.005389	3.559000	1.646300	0.000000	0.229	0.002 **
V6WTRR	0.017796	0.001927	9.234000	0.000000	1.525100	0.260	0.002 **
V46BHHE	0.017556	0.007155	2.454000	0.307900	1.798900	0.291	0.001 ***
V39FHE	0.017550	0.003257	5.388000	0.000000	1.532700	0.322	0.001 ***
V25WAG	0.016052	0.002282	7.033000	1.376000	0.000000	0.351	0.001 ***
V22SCR	0.015788	0.003354	4.707000	0.000000	1.350800	0.379	0.001 ***
V40BRTH	0.015511	0.009337	1.661000	0.000000	1.379300	0.406	0.004 **
V15STTH	0.015362	0.007523	2.042000	0.323400	1.627000	0.433	0.013 *
V44MUSK	0.013318	0.008639	1.542000	0.000000	1.180800	0.457	0.005 **
V580B0	0.012439	0.003104	4.007000	0.000000	1.059200	0.479	0.001 ***
V42SIL	0.012248	0.004880	2.510000	0.420400	1.255000	0.500	0.028 *
V75RCR	0.011824	0.003514	3.365000	0.000000	1.000900	0.521	0.001 ***
V28VS	0.011169	0.006772	1.649000	0.000000	0.971800	0.541	0.064 .
V11CR	0.011098	0.006986	1.589000	0.000000	0.992900	0.561	0.153
V33RWH	0.010578	0.006786	1.559000	0.579800	1.512200	0.580	0.038 *
V16LR	0.010525	0.007973	1.320000	1.184900	0.576500	0.598	0.102
V36F	0.010144	0.006630	1.530000	0.646500	1.330200	0.616	0.032 *
V29CST	0.009753	0.005485	1.778000	1.059100	0.256000	0.633	0.039 *
V23RBFT	0.009549	0.007039	1.357000	0.937700	0.320500	0.650	0.318
V43GCU	0.009226	0.005611	1.644000	0.000000	0.818400	0.667	0.058 .
V35STP	0.009032	0.007151	1.263000	0.993200	0.656700	0.683	0.229
V36YFHE	0.008694	0.006441	1.350000	1.243700	1.383500	0.698	0.625
V2EYR	0.008634	0.006941	1.244000	1.599500	0.874900	0.713	0.153
V56WH	0.008626	0.006596	1.308000	0.674800	1.171900	0.729	0.094 .
V65LWB	0.008297	0.008644	0.960000	0.741500	0.000000	0.743	0.145
V63DWS	0.008199	0.007646	1.072000	0.724800	0.220000	0.758	0.048 *
V91NMIN	0.008072	0.008448	0.955000	0.719700	0.000000	0.772	0.063 .
V21ESP	0.007818	0.008323	0.939000	0.644000	0.000000	0.786	0.681
V7WEHE	0.007692	0.006430	1.196000	1.241500	1.296700	0.799	0.331
V4BTH	0.007495	0.007712	0.972000	1.184200	1.520900	0.813	0.753
V56FTC	0.007373	0.007510	0.982000	0.615700	0.281200	0.826	0.303

V66CBW	0.007079	0.007358	0.962000	0.000000	0.566600	0.838	0.003	**
V90WTG	0.006660	0.007437	0.896000	0.000000	0.542000	0.850	0.037	*
V70RSL	0.006539	0.006415	1.019000	0.236400	0.557400	0.862	0.242	
V61SPW	0.006393	0.006678	0.957000	0.000000	0.593000	0.873	0.027	*
V19ER	0.006130	0.006577	0.932000	1.308600	0.846400	0.884	0.860	
V14AUR	0.005432	0.006537	0.831000	0.962100	1.211600	0.893	0.627	
V81SHBC	0.004610	0.006122	0.753000	0.271900	0.281200	0.902	0.940	
V12LK	0.004495	0.005039	0.892000	0.801000	1.122800	0.910	0.303	
V79YTHE	0.004087	0.007327	0.558000	0.000000	0.396100	0.917	0.052	.
V9SFW	0.004012	0.003799	1.056000	1.734700	1.472800	0.924	0.745	
V30BTR	0.003950	0.007083	0.558000	0.000000	0.382900	0.931	0.653	
V38GAL	0.003595	0.006455	0.557000	0.323400	0.000000	0.937	0.754	
V13RWB	0.003563	0.001691	2.107000	1.922100	1.792200	0.944	0.999	
V41SPP	0.003386	0.002315	1.463000	1.200500	1.440100	0.950	0.886	
V87LFC	0.003273	0.005878	0.557000	0.289600	0.000000	0.955	0.500	
V45MGLK	0.003023	0.005428	0.557000	0.271900	0.000000	0.961	0.764	
V8WNHE	0.002915	0.000912	3.194000	1.843200	1.595400	0.966	0.933	
V53MB	0.002779	0.004990	0.557000	0.250000	0.000000	0.971	0.763	
V94RBEE	0.002642	0.004736	0.558000	0.000000	0.256000	0.976	0.426	
V96DF	0.002642	0.004736	0.558000	0.000000	0.256000	0.980	0.400	
V76GBB	0.002548	0.004573	0.557000	0.000000	0.210200	0.985	0.422	
V16ST	0.002444	0.001755	1.393000	1.493300	1.425700	0.989	0.334	
V88WG	0.002395	0.004296	0.557000	0.000000	0.210200	0.993	0.392	
V95HBC	0.002395	0.004296	0.557000	0.000000	0.210200	0.997	0.051	.
V24BFCS	0.001441	0.001298	1.110000	1.112500	1.014400	1.000	0.985	
V20PCU	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V31AMAG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V32SCC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V37WHIP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V47RFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V48YTB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V49LYRE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V50CHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V51OWH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V52TRM	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V54STHR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V55LHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V57PINK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V59YR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V60LFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V62RBT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V64BELL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V67GGC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V68PIL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V69SKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V71PDV	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V72CRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V73JW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V74BCHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V77RRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V78LLOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V80RF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V82AZKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V83SFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V84YRTH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V85ROSE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V86BCO	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V89PCO	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V92NFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V93DB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V97PCL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V98FLAME	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V99WWT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V100WBWS	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V101LCOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	

V102KING 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma_River Red Gu

	average	sd	ratio	ava	avb	cumsum	p
V32SCC	0.025596	0.002660	9.624000	0.000000	2.225000	0.041 0.001	***
V13RWB	0.022078	0.001581	13.960000	1.922100	0.000000	0.077 0.001	***
V10WBSW	0.020296	0.002871	7.069000	1.760400	0.000000	0.110 0.001	***
V27NHHE	0.018894	0.005104	3.702000	1.646300	0.000000	0.140 0.002	**
V2EYR	0.018457	0.002843	6.492000	1.599500	0.000000	0.170 0.001	***
V26WWCH	0.018125	0.001284	14.119000	0.000000	1.578000	0.199 0.004	**
V8WNHE	0.018039	0.006464	2.791000	1.843200	0.289600	0.228 0.001	***
V30BTR	0.017484	0.001776	9.844000	0.000000	1.520300	0.257 0.001	***
V31AMAG	0.016036	0.001581	10.146000	0.000000	1.393900	0.283 0.002	**
V60LFB	0.015147	0.003696	4.098000	0.000000	1.312300	0.307 0.001	***
V38GAL	0.014698	0.007189	2.044000	0.323400	1.582800	0.331 0.002	**
V47RFC	0.014352	0.001420	10.106000	0.000000	1.251400	0.354 0.001	***
V36YFHE	0.014260	0.009360	1.524000	1.243700	0.000000	0.377 0.018	*
V7WEHE	0.013756	0.008586	1.602000	1.241500	0.000000	0.399 0.004	**
V41SPP	0.013744	0.002390	5.750000	1.200500	0.000000	0.421 0.003	**
V69SKF	0.013273	0.001217	10.902000	0.000000	1.156400	0.443 0.001	***
V4BTH	0.013108	0.008293	1.581000	1.184200	0.000000	0.464 0.071	.
V70RSL	0.011108	0.007818	1.421000	0.236400	1.085200	0.482 0.010	**
V45MGLK	0.010974	0.005854	1.875000	0.271900	1.216900	0.500 0.002	**
V98FLAME	0.010959	0.006960	1.575000	0.000000	0.964700	0.517 0.001	***
V73JW	0.010616	0.006635	1.600000	0.000000	0.934100	0.534 0.001	***
V34WSW	0.010143	0.007370	1.376000	1.856000	0.961700	0.551 0.158	
V52TRM	0.009996	0.006422	1.556000	0.000000	0.871300	0.567 0.012	*
V18YTH	0.009754	0.010215	0.955000	0.000000	0.845000	0.583 0.267	
V93DB	0.009507	0.005742	1.656000	0.000000	0.829800	0.598 0.001	***
V91NMIN	0.008898	0.008710	1.022000	0.719700	0.767400	0.612 0.024	*
V15STTH	0.008597	0.008559	1.004000	0.323400	0.745400	0.626 0.477	
V36F	0.008595	0.004954	1.735000	0.646500	1.077300	0.640 0.097	.
V6WTR	0.008498	0.008891	0.956000	0.000000	0.736100	0.654 0.328	
V94RBEE	0.008405	0.005103	1.647000	0.000000	0.732200	0.667 0.001	***
V65LWB	0.008183	0.008484	0.965000	0.741500	0.000000	0.681 0.150	
V23RBFT	0.008089	0.005959	1.357000	0.937700	0.456500	0.694 0.600	
V63DWS	0.008008	0.008299	0.965000	0.724800	0.000000	0.707 0.051	.
V28VS	0.007914	0.008215	0.963000	0.000000	0.686000	0.720 0.576	
V35STP	0.007822	0.007933	0.986000	0.993200	1.638300	0.732 0.582	
V56WH	0.007810	0.005625	1.388000	0.674800	0.765000	0.745 0.168	
V21ESP	0.007696	0.008143	0.945000	0.644000	0.000000	0.757 0.715	
V59YR	0.007520	0.007810	0.963000	0.000000	0.651800	0.769 0.001	***
V42SIL	0.007277	0.009602	0.758000	0.420400	0.442300	0.781 0.755	
V56FTC	0.007269	0.006882	1.056000	0.615700	0.220000	0.793 0.338	
V88WG	0.007075	0.007399	0.956000	0.000000	0.612900	0.804 0.002	**
V90WTG	0.007075	0.007399	0.956000	0.000000	0.612900	0.816 0.028	*
V99WWT	0.007063	0.007361	0.959000	0.000000	0.615300	0.827 0.001	***
V580B0	0.007005	0.007268	0.964000	0.000000	0.611700	0.838 0.115	
V9SFW	0.006866	0.003914	1.754000	1.734700	1.145500	0.849 0.316	
V33RWH	0.006745	0.006466	1.043000	0.579800	0.532700	0.860 0.625	
V20PCU	0.006711	0.007071	0.949000	0.000000	0.602800	0.871 0.510	
V16LR	0.006244	0.004942	1.264000	1.184900	1.252000	0.881 0.858	
V72CRP	0.006087	0.006321	0.963000	0.000000	0.545600	0.891 0.001	***
V92NFB	0.006062	0.006313	0.960000	0.000000	0.513100	0.901 0.007	**
V71PD0V	0.006011	0.006489	0.926000	0.000000	0.530500	0.910 0.001	***
V12LK	0.005937	0.005887	1.009000	0.801000	0.910300	0.920 0.057	.
V14AUR	0.005129	0.006478	0.792000	0.962100	1.237700	0.928 0.665	
V87LFC	0.005124	0.006716	0.763000	0.289600	0.314400	0.937 0.206	
V84YRTH	0.004535	0.008137	0.557000	0.000000	0.389600	0.944 0.045	*
V101LCOR	0.004040	0.007245	0.558000	0.000000	0.370000	0.951 0.001	***
V53MB	0.004022	0.005167	0.778000	0.250000	0.210200	0.957 0.594	
V29CST	0.003883	0.005041	0.770000	1.059100	0.776000	0.963 0.975	

V46BHHE	0.003418	0.006118	0.559000	0.307900	0.000000	0.969	0.990
V74BCHE	0.003144	0.005642	0.557000	0.000000	0.261700	0.974	0.002 **
V81SHBC	0.003030	0.005425	0.559000	0.271900	0.000000	0.979	0.987
V17WPHE	0.002749	0.001706	1.611000	2.247700	2.034900	0.983	0.982
V24BFCS	0.002646	0.002193	1.206000	1.112500	1.328700	0.988	0.904
V95HBC	0.002389	0.004287	0.557000	0.000000	0.198800	0.991	0.047 *
V16ST	0.002019	0.001634	1.236000	1.493300	1.343300	0.995	0.553
V19ER	0.001923	0.001172	1.640000	1.308600	1.397300	0.998	0.996
V25WAG	0.001422	0.001101	1.292000	1.376000	1.365200	1.000	0.959
V11CR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V22SCR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V37WHIP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V39FHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V40BRTH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V43GCU	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V44MUSK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V48YTBC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V49LYRE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V50CHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V51OWH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V54STHR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V55LHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V57PINK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V61SPW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V62RBTR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V64BELL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V66CBW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V67GGC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V68PIL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V75RCR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V76GBB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V77RRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V78LLOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V79YTHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V80RF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V82AZKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V83SFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V85ROSE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V86BC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V89PC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V96DF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V97PCL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V100WBWS	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V102KING	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Montane Fore_Foothills Wo

	average	sd	ratio	ava	avb	cumsum	p
V40BRTH	0.014435	0.009183	1.571800	0.000000	1.423800	0.042	0.015 *
V18YTH	0.012164	0.007560	1.609000	0.000000	1.199300	0.077	0.075 .
V26WWCH	0.010745	0.008003	1.342600	0.462400	1.210300	0.108	0.142
V22SCR	0.010509	0.005505	1.909200	0.317500	1.364500	0.138	0.030 *
V37WHIP	0.010114	0.006054	1.670700	1.020400	0.000000	0.168	0.021 *
V48YTBC	0.009915	0.005918	1.675400	0.958400	0.000000	0.196	0.027 *
V83SFC	0.009752	0.004877	1.999500	1.220000	0.261700	0.224	0.029 *
V10WBWS	0.009395	0.007408	1.268200	1.622600	0.773600	0.251	0.411
V67GGC	0.009162	0.005646	1.622600	0.926600	0.000000	0.278	0.006 **
V69SKF	0.009034	0.005447	1.658400	0.000000	0.890600	0.304	0.039 *
V53MB	0.008754	0.005324	1.644300	0.000000	0.851600	0.329	0.017 *
V33RWH	0.008585	0.007066	1.215100	0.753700	1.597500	0.354	0.271
V46BHHE	0.008556	0.006287	1.361000	0.828800	0.931700	0.379	0.534
V13RWB	0.008482	0.004846	1.750400	0.662400	1.170600	0.403	0.696
V49LYRE	0.007927	0.004849	1.634700	0.803100	0.000000	0.426	0.030 *

V68PIL	0.007799	0.004781	1.631300	0.785200	0.000000	0.449	0.013	*
V80RF	0.007785	0.006023	1.292600	0.913500	0.304500	0.471	0.122	
V54STHR	0.007684	0.004615	1.665200	0.758900	0.000000	0.493	0.004	**
V23RBFT	0.007670	0.005612	1.366800	0.883700	0.243500	0.516	0.704	
V28VS	0.007670	0.007611	1.007700	0.709200	1.088900	0.538	0.640	
V43GCU	0.007662	0.005636	1.359400	1.256100	0.583000	0.560	0.213	
V16LR	0.007478	0.006026	1.241100	1.023200	1.479800	0.582	0.662	
V20PCU	0.006977	0.006456	1.080800	1.268100	0.673000	0.602	0.372	
V19ER	0.006769	0.006493	1.042500	0.266900	0.654600	0.621	0.753	
V29CST	0.006698	0.006812	0.983200	0.314400	0.659900	0.641	0.487	
V50CHE	0.006550	0.006045	1.083500	0.509600	0.384600	0.659	0.465	
V35STP	0.006186	0.006202	0.997400	0.944200	1.011200	0.677	0.865	
V42SIL	0.006120	0.004742	1.290500	1.363600	0.761200	0.695	0.921	
V24BFCS	0.005843	0.005215	1.120300	0.496700	0.794400	0.712	0.413	
V14AUR	0.005747	0.006056	0.948900	0.919800	0.987000	0.729	0.580	
V55LHE	0.005677	0.005865	0.967900	0.590500	0.000000	0.745	0.262	
V36YFHE	0.005166	0.005585	0.925100	1.134500	1.319700	0.760	0.976	
V9SFW	0.005041	0.006178	0.815900	1.525000	1.081900	0.774	0.590	
V64BELL	0.004931	0.008845	0.557500	0.000000	0.492000	0.789	0.362	
V86BC00	0.004867	0.005040	0.965600	0.481700	0.000000	0.803	0.083	.
V81SHBC	0.004290	0.004697	0.913300	1.191300	0.807100	0.815	0.953	
V62RBTR	0.004175	0.005389	0.774800	0.281200	0.220000	0.827	0.182	
V21ESP	0.003642	0.002780	1.310000	1.502100	1.346100	0.838	0.985	
V96DF	0.003541	0.006353	0.557400	0.000000	0.349000	0.848	0.240	
V4BTH	0.003496	0.002575	1.358000	2.294000	1.975000	0.858	0.948	
V61SPW	0.003310	0.005938	0.557400	0.000000	0.326200	0.868	0.285	
V8WNHE	0.003301	0.002559	1.290000	1.709100	1.582200	0.877	0.897	
V85ROSE	0.003237	0.005792	0.558900	0.331700	0.000000	0.886	0.338	
V7WEHE	0.002775	0.001747	1.588400	1.224500	1.318200	0.894	0.944	
V27NHHE	0.002698	0.004842	0.557300	0.000000	0.256000	0.902	0.977	
V580B0	0.002685	0.004816	0.557400	0.000000	0.261700	0.910	0.870	
V11CR	0.002557	0.001689	1.513300	1.710400	1.484400	0.917	0.991	
V63DWS	0.002440	0.004377	0.557500	0.000000	0.243500	0.924	0.810	
V57PINK	0.002440	0.004365	0.558900	0.250000	0.000000	0.931	0.464	
V76GBB	0.002313	0.004139	0.558900	0.220000	0.000000	0.938	0.577	
V66CBW	0.002257	0.004050	0.557400	0.000000	0.220000	0.945	0.599	
V102KING	0.002240	0.004008	0.558900	0.236400	0.000000	0.951	0.334	
V41SPP	0.002181	0.001351	1.614700	1.352200	1.526000	0.958	0.946	
V510WH	0.002147	0.003842	0.558900	0.220000	0.000000	0.964	0.607	
V16ST	0.002076	0.002289	0.906700	1.379200	1.577800	0.970	0.524	
V2EYR	0.002039	0.001545	1.319900	1.440100	1.316800	0.976	0.960	
V15STTH	0.001903	0.001654	1.150800	1.857500	1.788200	0.981	0.990	
V56WH	0.001891	0.001382	1.367800	1.486100	1.309900	0.987	0.972	
V3GF	0.001741	0.001033	1.685600	1.724600	1.569600	0.992	0.991	
V6WTR	0.001138	0.000630	1.805600	1.567100	1.579900	0.995	0.995	
V56FTC	0.001005	0.000717	1.402700	1.231300	1.244700	0.998	0.998	
V12LK	0.000762	0.000683	1.116400	1.324100	1.350100	1.000	0.996	
V17WPHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V25WAG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V30BTR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V31AMAG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V32SCC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V34WSW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V38GAL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V39FHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V44MUSK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V45MGLK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V47RFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V52TRM	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V59YR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V60LFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V65LWB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V70RSL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V71PD0V	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V72CRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	

V73JW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V74BCHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V75RCR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V77RRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V78LLOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V79YTHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V82AZKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V84YRTH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V87LFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V88WG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V89PC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V90WTG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V91NM1N	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V92NFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V93DB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V94RBEE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V95HBC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V97PCL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V98FLAME	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V99WWT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V100WBWS	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V101LCOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Montane Fore_Box-Ironbark

	average	sd	ratio	ava	avb	cumsum	p
V18YTH	0.019614	0.002267	8.650000	0.000000	1.909900	0.042	0.002 **
V10WBSW	0.016737	0.002289	7.312000	1.622600	0.000000	0.079	0.005 **
V39FHE	0.015508	0.003007	5.157000	0.000000	1.532700	0.112	0.002 **
V21ESP	0.015277	0.002850	5.360000	1.502100	0.000000	0.145	0.005 **
V40BRTH	0.013739	0.008267	1.662000	0.000000	1.379300	0.175	0.012 *
V20PCU	0.013073	0.001670	7.829000	1.268100	0.000000	0.203	0.004 **
V26WWCH	0.012932	0.006735	1.920000	0.462400	1.655500	0.231	0.045 *
V13RWB	0.012500	0.008069	1.549000	0.662400	1.792200	0.258	0.070 .
V83SFC	0.012497	0.001221	10.232000	1.220000	0.000000	0.285	0.004 **
V44MUSK	0.011793	0.007641	1.543000	0.000000	1.180800	0.310	0.009 **
V580B0	0.010960	0.002648	4.139000	0.000000	1.059200	0.334	0.002 **
V22SCR	0.010674	0.005558	1.920000	0.317500	1.350800	0.357	0.026 *
V46BHHE	0.010527	0.009610	1.095000	0.828800	1.798900	0.380	0.136
V75RCR	0.010410	0.003002	3.468000	0.000000	1.000900	0.402	0.001 ***
V37WHIP	0.010161	0.006135	1.656000	1.020400	0.000000	0.424	0.020 *
V56FTC	0.010047	0.005497	1.828000	1.231300	0.281200	0.446	0.041 *
V48YTBC	0.009965	0.006005	1.659000	0.958400	0.000000	0.468	0.024 *
V81SHBC	0.009699	0.005338	1.817000	1.191300	0.281200	0.489	0.021 *
V67GGC	0.009203	0.005719	1.609000	0.926600	0.000000	0.508	0.002 **
V80RF	0.009127	0.005635	1.620000	0.913500	0.000000	0.528	0.057 .
V16LR	0.008206	0.006541	1.255000	1.023200	0.576500	0.546	0.494
V11CR	0.008012	0.007131	1.124000	1.710400	0.992900	0.563	0.762
V49LYRE	0.007963	0.004911	1.621000	0.803100	0.000000	0.581	0.020 *
V4BTH	0.007900	0.002729	2.895000	2.294000	1.520900	0.598	0.690
V33RWH	0.007855	0.007080	1.109000	0.753700	1.512200	0.615	0.365
V23RBFT	0.007852	0.005901	1.331000	0.883700	0.320500	0.631	0.686
V68PIL	0.007835	0.004844	1.617000	0.785200	0.000000	0.648	0.012 *
V35STP	0.007792	0.006378	1.222000	0.944200	0.656700	0.665	0.592
V54STHR	0.007721	0.004680	1.650000	0.758900	0.000000	0.682	0.004 **
V28VS	0.007362	0.006716	1.096000	0.709200	0.971800	0.698	0.760
V19ER	0.007289	0.005572	1.308000	0.266900	0.846400	0.714	0.616
V36YFHE	0.006921	0.005444	1.271000	1.134500	1.383500	0.729	0.887
V66CBW	0.006194	0.006437	0.962000	0.000000	0.566600	0.742	0.026 *
V24BFCS	0.005995	0.005034	1.191000	0.496700	1.014400	0.755	0.378
V2EYR	0.005841	0.005810	1.005000	1.440100	0.874900	0.768	0.608
V90WTG	0.005838	0.006497	0.899000	0.000000	0.542000	0.780	0.107
V70RSL	0.005780	0.006013	0.961000	0.000000	0.557400	0.793	0.357

V55LHE	0.005702	0.005915	0.964000	0.590500	0.000000	0.805	0.261
V61SPW	0.005691	0.005950	0.956000	0.000000	0.593000	0.817	0.054
V50CHE	0.005164	0.005411	0.954000	0.509600	0.000000	0.829	0.659
V86BC00	0.004890	0.005087	0.961000	0.481700	0.000000	0.839	0.104
V43GCU	0.004840	0.005651	0.857000	1.256100	0.818400	0.850	0.860
V29CST	0.004252	0.005499	0.773000	0.314400	0.256000	0.859	0.964
V14AUR	0.004172	0.005415	0.771000	0.919800	1.211600	0.868	0.851
V3GF	0.004016	0.002231	1.800000	1.724600	1.330200	0.876	0.669
V79YTHE	0.003656	0.006555	0.558000	0.000000	0.396100	0.884	0.191
V30BTR	0.003534	0.006336	0.558000	0.000000	0.382900	0.892	0.808
V76GBB	0.003414	0.004520	0.755000	0.220000	0.210200	0.899	0.272
V56WH	0.003383	0.002341	1.445000	1.486100	1.171900	0.907	0.803
V85ROSE	0.003252	0.005834	0.557000	0.331700	0.000000	0.914	0.345
V62RBTR	0.003172	0.005698	0.557000	0.281200	0.000000	0.920	0.368
V8WNHE	0.002763	0.002154	1.282000	1.709100	1.595400	0.926	0.941
V15STTH	0.002604	0.001970	1.321000	1.857500	1.627000	0.932	0.960
V57PINK	0.002450	0.004397	0.557000	0.250000	0.000000	0.937	0.493
V9SFW	0.002418	0.002010	1.203000	1.525000	1.472800	0.943	0.955
V94RBEE	0.002363	0.004237	0.558000	0.000000	0.256000	0.948	0.548
V96DF	0.002363	0.004237	0.558000	0.000000	0.256000	0.953	0.517
V102KING	0.002249	0.004035	0.557000	0.236400	0.000000	0.958	0.323
V7WEHE	0.002224	0.001655	1.344000	1.224500	1.296700	0.963	0.982
V510WH	0.002157	0.003870	0.557000	0.220000	0.000000	0.967	0.607
V88WG	0.002119	0.003801	0.557000	0.000000	0.210200	0.972	0.523
V95HBC	0.002119	0.003801	0.557000	0.000000	0.210200	0.976	0.325
V12LK	0.002086	0.000765	2.726000	1.324100	1.122800	0.981	0.627
V63DWS	0.002031	0.003641	0.558000	0.000000	0.220000	0.985	0.928
V42SIL	0.002007	0.001772	1.133000	1.363600	1.255000	0.990	1.000
V16ST	0.001927	0.000613	3.143000	1.379200	1.425700	0.994	0.610
V41SPP	0.001767	0.001227	1.439000	1.352200	1.440100	0.998	0.945
V6WTR	0.001142	0.000614	1.860000	1.567100	1.525100	1.000	0.992
V17WPHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V25WAG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V27NHHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V31AMAG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V32SCC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V34WSW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V38GAL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V45MGLK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V47RFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V52TRM	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V53MB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V59YR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V60LFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V64BELL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V65LWB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V69SKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V71PD0V	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V72CRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V73JW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V74BCHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V77RRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V78LLOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V82AZKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V84YRTH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V87LFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V89PC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V91NMIN	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V92NFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V93DB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V97PCL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V98FLAME	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V99WWT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V100WBWS	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V101LCOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Montane Fore_River Red Gu

	average	sd	ratio	ava	avb	cumsum	p
V4BTH	0.023240	0.001423	16.334000	2.294000	0.000000	0.032	0.001 ***
V32SCC	0.022619	0.002319	9.752000	0.000000	2.225000	0.062	0.002 **
V17WPHE	0.020648	0.001817	11.364000	0.000000	2.034900	0.090	0.007 **
V11CR	0.017360	0.001975	8.790000	1.710400	0.000000	0.114	0.004 **
V10WBSW	0.016529	0.001948	8.485000	1.622600	0.000000	0.136	0.005 **
V38GAL	0.016118	0.001999	8.065000	0.000000	1.582800	0.158	0.001 ***
V30BTR	0.015451	0.001552	9.955000	0.000000	1.520300	0.179	0.002 **
V21ESP	0.015090	0.002641	5.713000	1.502100	0.000000	0.200	0.003 **
V2EYR	0.014573	0.001010	14.428000	1.440100	0.000000	0.220	0.001 ***
V8WNHE	0.014442	0.006232	2.317000	1.709100	0.289600	0.239	0.003 **
V31AMAG	0.014171	0.001380	10.268000	0.000000	1.393900	0.258	0.002 **
V25WAG	0.013859	0.001151	12.039000	0.000000	1.365200	0.277	0.003 **
V41SPP	0.013725	0.001684	8.149000	1.352200	0.000000	0.296	0.002 **
V60LFB	0.013381	0.003234	4.138000	0.000000	1.312300	0.314	0.001 ***
V47RFC	0.012687	0.001289	9.845000	0.000000	1.251400	0.331	0.001 ***
V43GCU	0.012658	0.001321	9.581000	1.256100	0.000000	0.348	0.008 **
V7WEHE	0.012401	0.000859	14.431000	1.224500	0.000000	0.365	0.010 **
V26WWCH	0.012397	0.005637	2.199000	0.462400	1.578000	0.382	0.055 .
V83SFC	0.012343	0.000888	13.906000	1.220000	0.000000	0.399	0.002 **
V45MGLK	0.012337	0.001123	10.986000	0.000000	1.216900	0.416	0.001 ***
V81SHBC	0.012054	0.001019	11.835000	1.191300	0.000000	0.432	0.003 **
V69SKF	0.011732	0.001094	10.720000	0.000000	1.156400	0.448	0.001 ***
V42SIL	0.011523	0.004695	2.455000	1.363600	0.442300	0.464	0.063 .
V19ER	0.011415	0.005033	2.268000	0.266900	1.397300	0.479	0.025 *
V36YFHE	0.011410	0.006858	1.664000	1.134500	0.000000	0.495	0.127
V15STTH	0.011292	0.008100	1.394000	1.857500	0.745400	0.510	0.125
V70RSL	0.010957	0.007153	1.532000	0.000000	1.085200	0.525	0.016 *
V56FTC	0.010350	0.004225	2.450000	1.231300	0.220000	0.539	0.025 *
V37WHIP	0.010040	0.006016	1.669000	1.020400	0.000000	0.552	0.025 *
V48YBIC	0.009839	0.005880	1.673000	0.958400	0.000000	0.566	0.020 *
V34WSW	0.009813	0.006345	1.546000	0.000000	0.961700	0.579	0.193
V98FLAME	0.009699	0.006152	1.577000	0.000000	0.964700	0.592	0.010 **
V73JW	0.009395	0.005864	1.602000	0.000000	0.934100	0.605	0.001 ***
V67GGC	0.009094	0.005611	1.621000	0.926600	0.000000	0.618	0.005 **
V80RF	0.009018	0.005525	1.632000	0.913500	0.000000	0.630	0.066 .
V52TRM	0.008836	0.005669	1.559000	0.000000	0.871300	0.642	0.034 *
V6WTTR	0.008775	0.007475	1.174000	1.567100	0.736100	0.654	0.307
V18YTH	0.008617	0.009023	0.955000	0.000000	0.845000	0.665	0.502
V24BFCS	0.008611	0.005836	1.475000	0.496700	1.328700	0.677	0.027 *
V93DB	0.008405	0.005073	1.657000	0.000000	0.829800	0.689	0.003 **
V91NMIN	0.007934	0.008238	0.963000	0.000000	0.767400	0.699	0.074 .
V46BHHE	0.007914	0.008183	0.967000	0.828800	0.000000	0.710	0.667
V49LYRE	0.007869	0.004819	1.633000	0.803100	0.000000	0.721	0.032 *
V33RWH	0.007804	0.006307	1.237000	0.753700	0.532700	0.731	0.424
V68PIL	0.007742	0.004750	1.630000	0.785200	0.000000	0.742	0.024 *
V54STHR	0.007627	0.004585	1.663000	0.758900	0.000000	0.752	0.008 **
V20PCU	0.007523	0.006020	1.250000	1.268100	0.602800	0.762	0.192
V94RBEE	0.007429	0.004516	1.645000	0.000000	0.732200	0.772	0.005 **
V56WH	0.007231	0.004781	1.512000	1.486100	0.765000	0.782	0.256
V28VS	0.007207	0.007247	0.994000	0.709200	0.686000	0.792	0.766
V29CST	0.007165	0.004999	1.433000	0.314400	0.776000	0.802	0.307
V35STP	0.007050	0.006207	1.136000	0.944200	1.638300	0.811	0.761
V59YR	0.006644	0.006900	0.963000	0.000000	0.651800	0.820	0.020 *
V23RBFT	0.006615	0.005151	1.284000	0.883700	0.456500	0.829	0.895
V36F	0.006534	0.001821	3.589000	1.724600	1.077300	0.838	0.437
V13RWB	0.006293	0.007149	0.880000	0.662400	0.000000	0.847	0.951
V16LR	0.006278	0.005139	1.222000	1.023200	1.252000	0.855	0.830
V88WG	0.006250	0.006536	0.956000	0.000000	0.612900	0.864	0.022 *
V90WTG	0.006250	0.006536	0.956000	0.000000	0.612900	0.872	0.073 .

V99WWT	0.006243	0.006502	0.960000	0.000000	0.615300	0.881	0.024	*
V580B0	0.006193	0.006425	0.964000	0.000000	0.611700	0.889	0.234	
V55LHE	0.005637	0.005827	0.967000	0.590500	0.000000	0.897	0.253	
V72CRP	0.005399	0.005608	0.963000	0.000000	0.545600	0.904	0.018	*
V92NFB	0.005341	0.005564	0.960000	0.000000	0.513100	0.911	0.074	.
V71PD0V	0.005320	0.005756	0.924000	0.000000	0.530500	0.919	0.043	*
V50CHE	0.005101	0.005325	0.958000	0.509600	0.000000	0.926	0.638	
V86BC00	0.004831	0.005006	0.965000	0.481700	0.000000	0.932	0.106	
V9SFW	0.004284	0.002149	1.993000	1.525000	1.145500	0.938	0.687	
V12LK	0.004232	0.005626	0.752000	1.324100	0.910300	0.944	0.373	
V14AUR	0.004098	0.005269	0.778000	0.919800	1.237700	0.949	0.867	
V84YRTH	0.004003	0.007182	0.557000	0.000000	0.389600	0.955	0.209	
V101LCOR	0.003592	0.006442	0.558000	0.000000	0.370000	0.960	0.193	
V22SCR	0.003532	0.006323	0.559000	0.317500	0.000000	0.964	0.885	
V85ROSE	0.003214	0.005752	0.559000	0.331700	0.000000	0.969	0.382	
V62RBTR	0.003128	0.005600	0.559000	0.281200	0.000000	0.973	0.382	
V87LFC	0.003051	0.005472	0.558000	0.000000	0.314400	0.977	0.606	
V74BCHE	0.002765	0.004961	0.557000	0.000000	0.261700	0.981	0.180	
V57PINK	0.002422	0.004335	0.559000	0.250000	0.000000	0.984	0.508	
V76GBB	0.002296	0.004109	0.559000	0.220000	0.000000	0.987	0.594	
V102KING	0.002224	0.003981	0.559000	0.236400	0.000000	0.990	0.315	
V53MB	0.002160	0.003875	0.557000	0.000000	0.210200	0.993	0.874	
V510WH	0.002132	0.003815	0.559000	0.220000	0.000000	0.996	0.645	
V95HBC	0.002101	0.003770	0.557000	0.000000	0.198800	0.999	0.348	
V16ST	0.000715	0.000313	2.283000	1.379200	1.343300	1.000	0.999	
V27NHHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V39FHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V40BRTH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V44MUSK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V61SPW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V63DWS	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V64BELL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V65LWB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V66CBW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V75RCR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V77RRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V78LLOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V79YTHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V82AZKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V89PC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V96DF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V97PCL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	
V100WBWS	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Foothills Wo_Box-Ironbark

	average	sd	ratio	ava	avb	cumsum	p
V39FHE	0.015833	0.002871	5.514000	0.000000	1.532700	0.047	0.002 **
V21ESP	0.014135	0.003111	4.543000	1.346100	0.000000	0.089	0.006 **
V44MUSK	0.012037	0.007747	1.554000	0.000000	1.180800	0.125	0.013 *
V75RCR	0.010634	0.002985	3.562000	0.000000	1.000900	0.157	0.001 ***
V56FTC	0.010423	0.005633	1.850000	1.244700	0.281200	0.188	0.033 *
V16LR	0.009727	0.006963	1.397000	1.479800	0.576500	0.217	0.182
V40BRTH	0.009298	0.009205	1.010000	1.423800	1.379300	0.245	0.195
V69SKF	0.009269	0.005598	1.656000	0.890600	0.000000	0.273	0.016 *
V46BHHE	0.009105	0.006329	1.439000	0.931700	1.798900	0.300	0.362
V53MB	0.008985	0.005474	1.641000	0.851600	0.000000	0.327	0.021 *
V580B0	0.008774	0.004921	1.783000	0.261700	1.059200	0.353	0.029 *
V35STP	0.008296	0.006636	1.250000	1.011200	0.656700	0.378	0.427
V10WBSW	0.008277	0.008818	0.939000	0.773600	0.000000	0.402	0.615
V18YTH	0.007632	0.008144	0.937000	1.199300	1.909900	0.425	0.590
V81SHBC	0.007489	0.005740	1.305000	0.807100	0.281200	0.447	0.189
V20PCU	0.006999	0.007268	0.963000	0.673000	0.000000	0.468	0.344

V19ER	0.006999	0.005848	1.197000	0.654600	0.846400	0.489	0.703
V29CST	0.006862	0.006603	1.039000	0.659900	0.256000	0.510	0.432
V43GCU	0.006594	0.005495	1.200000	0.583000	0.818400	0.529	0.496
V13RWB	0.006454	0.003518	1.834000	1.170600	1.792200	0.549	0.938
V11CR	0.006420	0.006540	0.982000	1.484400	0.992900	0.568	0.922
V28VS	0.006385	0.006620	0.964000	1.088900	0.971800	0.587	0.885
V61SPW	0.006374	0.006302	1.011000	0.326200	0.593000	0.606	0.020 *
V66CBW	0.006186	0.005875	1.053000	0.220000	0.566600	0.624	0.022 *
V26WWCH	0.006051	0.007359	0.822000	1.210300	1.655500	0.642	0.869
V90WTG	0.005966	0.006614	0.902000	0.000000	0.542000	0.660	0.105
V70RSL	0.005903	0.006116	0.965000	0.000000	0.557400	0.678	0.303
V9SFW	0.005785	0.005795	0.998000	1.081900	1.472800	0.695	0.465
V42SIL	0.005356	0.004742	1.130000	0.761200	1.255000	0.711	0.956
V2EYR	0.005255	0.005672	0.927000	1.316800	0.874900	0.727	0.745
V64BELL	0.005057	0.009076	0.557000	0.492000	0.000000	0.742	0.319
V4BTH	0.005028	0.003157	1.593000	1.975000	1.520900	0.757	0.852
V96DF	0.004848	0.006302	0.769000	0.349000	0.256000	0.771	0.071
V23RBFT	0.004574	0.005837	0.784000	0.243500	0.320500	0.785	0.983
V36YFHE	0.004518	0.002675	1.689000	1.319700	1.383500	0.798	0.981
V14AUR	0.004445	0.005204	0.854000	0.987000	1.211600	0.812	0.800
V50CHE	0.004163	0.007474	0.557000	0.384600	0.000000	0.824	0.794
V79YTHE	0.003726	0.006667	0.559000	0.000000	0.396100	0.835	0.173
V30BTR	0.003602	0.006445	0.559000	0.000000	0.382900	0.846	0.776
V63DWS	0.003558	0.004682	0.760000	0.243500	0.220000	0.857	0.674
V24BFCS	0.003325	0.004364	0.762000	0.794400	1.014400	0.867	0.893
V80RF	0.003296	0.005917	0.557000	0.304500	0.000000	0.876	0.902
V7WEHE	0.003045	0.002323	1.311000	1.318200	1.296700	0.885	0.911
V16ST	0.002847	0.001972	1.444000	1.577800	1.425700	0.894	0.145
V27NHHE	0.002771	0.004976	0.557000	0.256000	0.000000	0.902	0.967
V83SFC	0.002755	0.004946	0.557000	0.261700	0.000000	0.910	0.905
V15STTH	0.002694	0.001759	1.532000	1.788200	1.627000	0.918	0.959
V36F	0.002586	0.002274	1.137000	1.569600	1.330200	0.926	0.944
V94RBEE	0.002409	0.004310	0.559000	0.000000	0.256000	0.933	0.538
V12LK	0.002400	0.001272	1.887000	1.350100	1.122800	0.941	0.547
V62RBTR	0.002317	0.004159	0.557000	0.220000	0.000000	0.947	0.593
V8WNHE	0.002311	0.000652	3.548000	1.582200	1.595400	0.954	0.981
V76GBB	0.002287	0.004092	0.559000	0.000000	0.210200	0.961	0.613
V22SCR	0.002258	0.001532	1.474000	1.364500	1.350800	0.968	0.971
V56WH	0.002206	0.001744	1.265000	1.309900	1.171900	0.975	0.937
V88WG	0.002163	0.003870	0.559000	0.000000	0.210200	0.981	0.484
V95HBC	0.002163	0.003870	0.559000	0.000000	0.210200	0.987	0.257
V33RWH	0.001840	0.001210	1.520000	1.597500	1.512200	0.993	0.996
V41SPP	0.001677	0.001086	1.544000	1.526000	1.440100	0.998	0.957
V6WTTR	0.000709	0.000533	1.332000	1.579900	1.525100	1.000	0.999
V17WPHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V25WAG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V31AMAG	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V32SCC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V34WSW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V37WHIP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V38GAL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V45MGLK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V47RFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V48YTBC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V49LYRE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V51OWH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V52TRM	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V54STHR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V55LHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V57PINK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V59YR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V60LFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V65LWB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V67GGC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V68PIL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

V71PDOV	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V72CRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V73JW	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V74BCHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V77RRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V78LLOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V82AZKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V84YRTH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V85ROSE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V86BC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V87LFC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V89PC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V91NMIN	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V92NFB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V93DB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V97PCL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V98FLAME	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V99WWT	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V100WBWS	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V101LCOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V102KING	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Foothills Wo_River Red Gu

	average	sd	ratio	ava	avb	cumsum	p
V32SCC	0.023093	0.001828	12.633000	0.000000	2.225000	0.035 0.001	***
V17WPHE	0.021080	0.001241	16.985000	0.000000	2.034900	0.067 0.005	**
V4BTH	0.020489	0.002659	7.707000	1.975000	0.000000	0.098 0.002	**
V38GAL	0.016457	0.001736	9.477000	0.000000	1.582800	0.123 0.001	***
V41SPP	0.015828	0.001134	13.956000	1.526000	0.000000	0.146 0.001	***
V30BTR	0.015775	0.001205	13.087000	0.000000	1.520300	0.170 0.001	***
V11CR	0.015410	0.001314	11.727000	1.484400	0.000000	0.194 0.005	**
V40BRTH	0.014624	0.009247	1.582000	1.423800	0.000000	0.216 0.007	**
V31AMAG	0.014468	0.001047	13.821000	0.000000	1.393900	0.238 0.001	***
V25WAG	0.014149	0.000727	19.475000	0.000000	1.365200	0.259 0.004	**
V22SCR	0.014138	0.001253	11.280000	1.364500	0.000000	0.280 0.003	**
V21ESP	0.013956	0.002919	4.781000	1.346100	0.000000	0.301 0.014	*
V7WEHE	0.013692	0.003128	4.377000	1.318200	0.000000	0.322 0.002	**
V2EYR	0.013677	0.002086	6.556000	1.316800	0.000000	0.343 0.002	**
V60LFB	0.013662	0.003176	4.302000	0.000000	1.312300	0.363 0.001	***
V36YFHE	0.013654	0.002185	6.250000	1.319700	0.000000	0.384 0.024	*
V8WNHE	0.013515	0.005982	2.259000	1.582200	0.289600	0.404 0.009	**
V47RFC	0.012953	0.001004	12.896000	0.000000	1.251400	0.424 0.001	***
V45MGLK	0.012595	0.000795	15.834000	0.000000	1.216900	0.443 0.001	***
V13RWB	0.012138	0.001836	6.609000	1.170600	0.000000	0.461 0.085	.
V70RSL	0.011186	0.007252	1.543000	0.000000	1.085200	0.478 0.006	**
V15STTH	0.011042	0.008000	1.380000	1.788200	0.745400	0.495 0.180	
V33RWH	0.011004	0.005906	1.863000	1.597500	0.532700	0.511 0.031	*
V56FTC	0.010724	0.004330	2.477000	1.244700	0.220000	0.528 0.017	*
V34WSW	0.010020	0.006434	1.557000	0.000000	0.961700	0.543 0.205	
V98FLAME	0.009901	0.006236	1.588000	0.000000	0.964700	0.558 0.008	**
V46BHHE	0.009696	0.006173	1.571000	0.931700	0.000000	0.573 0.242	
V73JW	0.009590	0.005943	1.614000	0.000000	0.934100	0.587 0.001	***
V18YTH	0.009296	0.008040	1.156000	1.199300	0.845000	0.601 0.371	
V52TRM	0.009021	0.005747	1.570000	0.000000	0.871300	0.615 0.028	*
V6WTTT	0.008899	0.007759	1.147000	1.579900	0.736100	0.628 0.297	
V93DB	0.008581	0.005139	1.670000	0.000000	0.829800	0.641 0.001	***
V42SIL	0.008525	0.004802	1.776000	0.761200	0.442300	0.654 0.468	
V81SHBC	0.008439	0.005218	1.617000	0.807100	0.000000	0.667 0.081	.
V10WBSW	0.008169	0.008666	0.943000	0.773600	0.000000	0.679 0.660	
V91NMIN	0.008103	0.008379	0.967000	0.000000	0.767400	0.691 0.076	.
V19ER	0.007916	0.006665	1.188000	0.654600	1.397300	0.703 0.452	
V53MB	0.007757	0.005181	1.497000	0.851600	0.210200	0.715 0.056	.

V28VS	0.007735	0.007384	1.048000	1.088900	0.686000	0.727	0.634
V94RBEE	0.007585	0.004574	1.658000	0.000000	0.732200	0.738	0.003 **
V20PCU	0.007072	0.006526	1.084000	0.673000	0.602800	0.749	0.323
V29CST	0.006866	0.005049	1.360000	0.659900	0.776000	0.759	0.472
V59YR	0.006784	0.007017	0.967000	0.000000	0.651800	0.769	0.006 **
V35STP	0.006652	0.006625	1.004000	1.011200	1.638300	0.779	0.801
V88WG	0.006382	0.006647	0.960000	0.000000	0.612900	0.789	0.007 **
V90WTG	0.006382	0.006647	0.960000	0.000000	0.612900	0.799	0.052 .
V99WWT	0.006373	0.006612	0.964000	0.000000	0.615300	0.808	0.010 **
V580B0	0.006330	0.006124	1.034000	0.261700	0.611700	0.818	0.235
V26WWCH	0.005950	0.006536	0.910000	1.210300	1.578000	0.827	0.887
V43GCU	0.005949	0.006303	0.944000	0.583000	0.000000	0.836	0.703
V9SFW	0.005874	0.004177	1.406000	1.081900	1.145500	0.845	0.495
V5GWH	0.005644	0.004760	1.186000	1.309900	0.765000	0.853	0.550
V72CRP	0.005509	0.005702	0.966000	0.000000	0.545600	0.862	0.014 *
V24BFCS	0.005494	0.005150	1.067000	0.794400	1.328700	0.870	0.493
V92NFB	0.005455	0.005659	0.964000	0.000000	0.513100	0.878	0.090 .
V71PD0V	0.005431	0.005852	0.928000	0.000000	0.530500	0.886	0.024 *
V36F	0.005121	0.001989	2.575000	1.569600	1.077300	0.894	0.523
V64BELL	0.004995	0.008940	0.559000	0.492000	0.000000	0.902	0.348
V23RBFT	0.004889	0.004924	0.993000	0.243500	0.456500	0.909	0.984
V12LK	0.004650	0.005759	0.808000	1.350100	0.910300	0.916	0.289
V14AUR	0.004157	0.005215	0.797000	0.987000	1.237700	0.922	0.846
V50CHE	0.004108	0.007354	0.559000	0.384600	0.000000	0.928	0.793
V84YRTH	0.004088	0.007314	0.559000	0.000000	0.389600	0.935	0.166
V69SKF	0.003666	0.005226	0.701000	0.890600	1.156400	0.940	0.807
V101LCOR	0.003664	0.006556	0.559000	0.000000	0.370000	0.946	0.139
V96DF	0.003588	0.006422	0.559000	0.349000	0.000000	0.951	0.225
V61SPW	0.003353	0.006002	0.559000	0.326200	0.000000	0.956	0.219
V80RF	0.003252	0.005822	0.559000	0.304500	0.000000	0.961	0.908
V87LFC	0.003113	0.005569	0.559000	0.000000	0.314400	0.966	0.641
V74BCHE	0.002825	0.005054	0.559000	0.000000	0.261700	0.970	0.146
V27NHHE	0.002735	0.004896	0.559000	0.256000	0.000000	0.974	0.968
V83SFC	0.002720	0.004869	0.559000	0.261700	0.000000	0.978	0.914
V16LR	0.002626	0.002077	1.264000	1.479800	1.252000	0.982	0.987
V16ST	0.002484	0.002388	1.040000	1.577800	1.343300	0.986	0.309
V63DWS	0.002472	0.004425	0.559000	0.243500	0.000000	0.990	0.803
V62RBTR	0.002287	0.004094	0.559000	0.220000	0.000000	0.993	0.599
V66CBW	0.002287	0.004094	0.559000	0.220000	0.000000	0.997	0.642
V95HBC	0.002146	0.003840	0.559000	0.000000	0.198800	1.000	0.286
V37WHIP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V39FHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V44MUSK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V48YTB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V49LYRE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V510WH	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V54STHR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V55LHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V57PINK	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V65LWB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V67GGC	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V68PIL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V75RCR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V76GBB	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V77RRP	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V78LLOR	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V79YTHE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V82AZKF	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V85ROSE	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V86BC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V89PC00	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V97PCL	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V100WBWS	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA
V102KING	0.000000	0.000000	NaN	0.000000	0.000000	1.000	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Contrast: Box-Ironbark_River Red Gu

	average	sd	ratio	ava	avb	cumsum	p
V32SCC	0.023209	0.002485	9.340000	0.000000	2.225000	0.036 0.003	**
V17WPHE	0.021185	0.001961	10.802000	0.000000	2.034900	0.068 0.008	**
V46BHHE	0.018779	0.002092	8.977000	1.798900	0.000000	0.097 0.001	***
V13RWB	0.018603	0.002860	6.505000	1.792200	0.000000	0.126 0.001	***
V38GAL	0.016539	0.002119	7.804000	0.000000	1.582800	0.151 0.001	***
V4BTH	0.015839	0.002532	6.255000	1.520900	0.000000	0.175 0.014	*
V39FHE	0.015712	0.002885	5.447000	1.532700	0.000000	0.200 0.002	**
V41SPP	0.014960	0.001224	12.222000	1.440100	0.000000	0.222 0.001	***
V31AMAG	0.014541	0.001484	9.795000	0.000000	1.393900	0.245 0.003	**
V36YFHE	0.014477	0.004922	2.941000	1.383500	0.000000	0.267 0.012	*
V25WAG	0.014220	0.001253	11.353000	0.000000	1.365200	0.289 0.004	**
V22SCR	0.014102	0.002784	5.066000	1.350800	0.000000	0.311 0.002	**
V40BRTH	0.013917	0.008320	1.673000	1.379300	0.000000	0.332 0.010	**
V8WNHE	0.013741	0.005715	2.405000	1.595400	0.289600	0.353 0.010	**
V60LFB	0.013731	0.003352	4.097000	0.000000	1.312300	0.374 0.001	***
V7WEHE	0.013625	0.003185	4.278000	1.296700	0.000000	0.395 0.003	**
V47RFC	0.013017	0.001371	9.494000	0.000000	1.251400	0.415 0.002	**
V45MGLK	0.012658	0.001205	10.507000	0.000000	1.216900	0.435 0.002	**
V30BTR	0.012531	0.007039	1.780000	0.382900	1.520300	0.454 0.009	**
V69SKF	0.012038	0.001175	10.245000	0.000000	1.156400	0.472 0.002	**
V44MUSK	0.011946	0.007696	1.552000	1.180800	0.000000	0.491 0.006	**
V18YTH	0.011253	0.009055	1.243000	1.909900	0.845000	0.508 0.097	.
V42SIL	0.011190	0.003985	2.808000	1.255000	0.442300	0.525 0.080	.
V35STP	0.010880	0.007777	1.399000	0.656700	1.638300	0.542 0.072	.
V75RCR	0.010550	0.002971	3.552000	1.000900	0.000000	0.558 0.001	***
V33RWH	0.010203	0.005935	1.719000	1.512200	0.532700	0.574 0.052	.
V34WSW	0.010070	0.006524	1.544000	0.000000	0.961700	0.589 0.168	.
V11CR	0.009963	0.006247	1.595000	0.992900	0.000000	0.605 0.384	.
V98FLAME	0.009949	0.006321	1.574000	0.000000	0.964700	0.620 0.009	**
V73JW	0.009637	0.006026	1.599000	0.000000	0.934100	0.635 0.002	**
V15STTH	0.009594	0.007905	1.214000	1.627000	0.745400	0.649 0.392	.
V52TRM	0.009066	0.005827	1.556000	0.000000	0.871300	0.663 0.018	*
V2EYR	0.009033	0.005592	1.615000	0.874900	0.000000	0.677 0.126	.
V93DB	0.008624	0.005214	1.654000	0.000000	0.829800	0.690 0.001	***
V70RSL	0.008584	0.006538	1.313000	0.557400	1.085200	0.704 0.059	.
V6WTR	0.008576	0.007595	1.129000	1.525100	0.736100	0.717 0.351	.
V43GCU	0.008275	0.004993	1.657000	0.818400	0.000000	0.730 0.122	.
V91NMIN	0.008145	0.008464	0.962000	0.000000	0.767400	0.742 0.064	.
V16LR	0.007429	0.006376	1.165000	0.576500	1.252000	0.753 0.645	.
V28VS	0.007148	0.006803	1.051000	0.971800	0.686000	0.764 0.780	.
V29CST	0.006978	0.005342	1.306000	0.256000	0.776000	0.775 0.392	.
V59YR	0.006818	0.007086	0.962000	0.000000	0.651800	0.786 0.011	*
V580B0	0.006742	0.004954	1.361000	1.059200	0.611700	0.796 0.160	.
V90WTG	0.006683	0.006034	1.107000	0.542000	0.612900	0.806 0.039	*
V94RBEE	0.006671	0.005063	1.318000	0.256000	0.732200	0.817 0.011	*
V88WG	0.006408	0.005997	1.069000	0.210200	0.612900	0.826 0.011	*
V99WWT	0.006405	0.006678	0.959000	0.000000	0.615300	0.836 0.010	**
V66CBW	0.006280	0.006500	0.966000	0.566600	0.000000	0.846 0.029	*
V19ER	0.006192	0.005991	1.034000	0.846400	1.397300	0.855 0.827	.
V20PCU	0.006104	0.006440	0.948000	0.000000	0.602800	0.865 0.675	.
V61SPW	0.005762	0.006007	0.959000	0.593000	0.000000	0.874 0.038	*
V23RBFT	0.005699	0.005229	1.090000	0.320500	0.456500	0.882 0.954	.
V72CRP	0.005535	0.005754	0.962000	0.000000	0.545600	0.891 0.012	*
V92NFB	0.005483	0.005717	0.959000	0.000000	0.513100	0.899 0.083	.
V71PDOV	0.005457	0.005906	0.924000	0.000000	0.530500	0.908 0.026	*
V56WH	0.004967	0.004736	1.049000	1.171900	0.765000	0.915 0.691	.
V9SFW	0.004265	0.002744	1.554000	1.472800	1.145500	0.922 0.699	.
V84YRTH	0.004108	0.007375	0.557000	0.000000	0.389600	0.928 0.162	.
V56FTC	0.003830	0.004875	0.786000	0.281200	0.220000	0.934 0.969	.
V12LK	0.003782	0.004861	0.778000	1.122800	0.910300	0.940 0.417	.

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V79YTHE 0.003700 0.006623 0.559000 0.396100 0.000000 0.946 0.177
V101LCOR 0.003681 0.006605 0.557000 0.000000 0.370000 0.951 0.119
V24BFCS 0.003288 0.002264 1.452000 1.014400 1.328700 0.956 0.880
V95HBC 0.003249 0.004283 0.759000 0.210200 0.198800 0.961 0.039 *
V36F 0.003169 0.001979 1.601000 1.330200 1.077300 0.966 0.857
V87LFC 0.003127 0.005611 0.557000 0.000000 0.314400 0.971 0.598
V74BCHE 0.002840 0.005099 0.557000 0.000000 0.261700 0.975 0.141
V81SHBC 0.002627 0.004702 0.559000 0.281200 0.000000 0.979 0.995
V96DF 0.002392 0.004281 0.559000 0.256000 0.000000 0.983 0.509
V76GBB 0.002269 0.004061 0.559000 0.210200 0.000000 0.986 0.595
V53MB 0.002217 0.003979 0.557000 0.000000 0.210200 0.990 0.880
V63DWS 0.002056 0.003679 0.559000 0.220000 0.000000 0.993 0.931
V16ST 0.001950 0.001137 1.714000 1.425700 1.343300 0.996 0.596
V26WWCH 0.001348 0.001039 1.297000 1.655500 1.578000 0.998 0.987
V14AUR 0.001248 0.000601 2.111000 1.211600 1.237700 1.000 0.993
V10WBSW 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V21ESP 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V27NHHE 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V37WHIP 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V48YBTC 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V49LYRE 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V50CHE 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V51OWH 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V54STHR 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V55LHE 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V57PINK 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V62RBTB 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V64BELL 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V65LWB 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V67GGC 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V68PIL 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V77RRP 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V78LLOR 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V80RF 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V82AZKF 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V83SFC 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V85ROSE 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V86BC00 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V89PC00 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V97PCL 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V100WBWS 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
V102KING 0.000000 0.000000 NaN 0.000000 0.000000 1.000 NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Permutation: free
Number of permutations: 999

```

Extract one comparison

The output can be long. The following code extracts the first pairwise habitat comparison.

CODE

```

first_comparison ← summary(Birdsim)[[1]]
head(first_comparison, 10)

```

OUTPUT

```

      average      sd    ratio    ava    avb    cumsum    p
V17WPHE 0.01977874 0.011557166 1.711384 0.6019704 2.2477343 0.04046870 0.003
V34WSW  0.01897905 0.007112357 2.668462 0.2475148 1.8559931 0.07930117 0.001

```

V11CR	0.01685036	0.007496274	2.247831	1.4434071	0.0000000	0.11377819	0.001
V6WTR	0.01545551	0.006359050	2.430475	1.3258152	0.0000000	0.14540126	0.002
V15STTH	0.01493983	0.009011889	1.657791	1.4754269	0.3233922	0.17596921	0.008
V25WAG	0.01396604	0.005801616	2.407268	0.2050889	1.3759763	0.20454472	0.002
V27NHHE	0.01276927	0.008435386	1.513774	0.6891770	1.6462619	0.23067153	0.044
V36F	0.01248274	0.008487765	1.470674	1.5693228	0.6464682	0.25621210	0.001
V4BTH	0.01147159	0.009943672	1.153658	1.9464092	1.1842313	0.27968378	0.129
V36YFHE	0.01070821	0.008005858	1.337547	1.1213827	1.2437450	0.30159353	0.086

Interpretation

Species with larger average contributions are those that most strongly separate two habitat types.

These may be habitat-associated species, but SIMPER should be interpreted carefully because common and abundant species often contribute strongly.

Questions for students

Question 1

What does it mean if two sites are close together in the nMDS plot?

Answer guide

They have similar bird community composition based on Bray-Curtis dissimilarity.

Question 2

Why did we transform the bird abundance data before calculating Bray-Curtis dissimilarities?

Answer guide

The transformation reduces the dominance of very abundant species and allows less abundant species to contribute more to the multivariate pattern.

Question 3

What does a significant PERMANOVA tell us?

Answer guide

It tells us that bird community composition differs among habitat types more than expected by chance.

Question 4

Why should we check dispersion with `betadisper()`?

Answer guide

Because PERMANOVA can be influenced by differences in the spread of groups, not only differences in their centres.

Question 5

What does SIMPER add to the analysis?

Answer guide

SIMPER helps identify which species contribute most to the observed differences among habitat types.

Key take-home message

Hierarchical clustering, nMDS, ANOSIM, PERMANOVA and SIMPER all use the same basic idea: sites can be compared based on their whole community composition.

For the Mac Nally bird data, these tools let us ask whether forest and woodland habitat types support distinct bird assemblages, and which species contribute most to those differences.